

Liquidity Premia and Transaction Costs

Extended Abstract

Although transaction costs can alter optimal trading strategies significantly, standard literature (e.g., Constantinides (1986) among others) concludes that they only have a *second-order* effect on liquidity premia (i.e., the maximum expected return an investor is willing to exchange for a zero transaction cost). We show that this conclusion depends crucially on the assumption of a constant investment opportunity set.

In a regime-switching setting where the investment opportunity set (such as expected return, volatility, and liquidity) varies stochastically, we explicitly characterize the optimal trading strategy in terms of two coupled ordinary differential equations with free boundaries; we prove a verification theorem for the obtained solution; and we provide an extensive analysis of the optimal trading strategy and liquidity premium. We find that contrary to the standard finding, transaction costs can have a *first-order* effect on liquidity premia. This result follows from a simple but fundamental intuition that if the investment opportunity set changes stochastically, the optimal stock investment target in the no-transaction-cost case also changes stochastically and an investor optimally rebalances more often to avoid being too far away from the moving target, which in turn results in the transaction cost having a greater effect.

This result suggests that the presence of transaction costs can produce a significantly larger portion of the observed high risk premium than the literature on portfolio choice with a constant opportunity set and transaction costs indicates. However, we show that with reasonably calibrated parameters, the presence of transaction costs alone still cannot *fully* explain the magnitude of the equity premium puzzle, even when the investment opportunity set varies stochastically.

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Transaction costs are prevalent in almost all financial markets. Extensive research has been conducted on the optimal consumption and investment policy in the presence of transaction costs (e.g., Constantinides (1986), Vayanos (1998), Koo (1992), Liu and Loewenstein (2002), Liu (2004)). The presence of transaction costs significantly changes optimal consumption and investment strategies. For example, in the presence of transaction costs, continuous trade incurs infinite transaction costs, and thus even a small transaction cost can dramatically decrease the frequency of trade. However, most studies find the utility loss due to the presence of transaction costs to be small. In particular, Constantinides (1986) finds that the liquidity premium (i.e., the maximum expected return an investor is willing to exchange for a zero transaction cost) is small relative to the transaction cost, even for a suboptimal trading strategy, and thus concludes that transaction costs only have a second-order effect for asset pricing. In particular, in this framework, it seems unlikely that transaction costs can have an important role in understanding the cross section patterns of expected returns, the equity premium puzzle, or the small stock risk premia. This finding contrasts with many empirical studies that suggest the importance of transaction costs or related measures such as turnover in influencing the cross-section patterns of expected returns.¹

One of the common assumptions of the existing literature on optimal consumption and investment with transaction costs is that the investment opportunity set is constant. For example, Constantinides (1986), Vayanos (1998), Liu and Loewenstein (2002), and Liu (2004) all assume that not only the expected stock return and the return volatility, but also the liquidity (transaction costs) are constant throughout the investment horizon. Empirical research, however, documents a great deal of evidence against the constant investment opportunity set hypothesis. For example, Campbell (1991) and Lewellen (2003) find that expected returns on equities change over time. Schwert (1989) and Campbell and Hentschel

¹See, for example, Eleswarapu (1997) and Brennan, Chordia, and Subahmanyam (1998).

(1992) conclude that the volatilities of stock returns also vary substantially over time. In addition, the existence of large liquidity shocks (e.g., the 1987 crash, the 1998 LTCM event) suggests that transaction costs can also vary over time.

Taking into account the time-varying nature of the investment opportunity set may qualitatively change the conclusion in the standard literature (e.g., Constantinides (1986)) that transaction costs only have a second-order effect. Intuitively, the presence of transaction costs reduce an investor's welfare through two venues. One is through the transaction cost payment, and the other is through the deviation from the optimal strategy in the no-transaction-cost case. If investment-opportunity-set parameters change over time, the optimal stock investment target in the no-transaction-cost case would also change over time. Therefore as market conditions change through time, compared to the constant-investment-opportunity-set case, an investor would rebalance more often to avoid being too far away from the target if the transaction cost rate is small; thus the relative impact of the transaction cost would increase, mainly because of a greater transaction cost payment resulted from more frequent trading. In addition, if liquidity can vary dramatically through time, an investor would significantly change his trading strategy to minimize the loss from a bad liquidity shock, and thus the impact of transaction costs would also significantly increase, mainly due to the substantial deviation from the optimal strategy in the no-transaction-cost case. To quantify this, we build a model similar to that of Constantinides (1986) and Davis and Norman (1990), but with regime switching for fundamental parameters. Specifically, we consider the optimal consumption and investment problem for a small investor (i.e., one with no price impact) who derives constant relative risk averse (CRRA) utility from intertemporal consumption and bequest.² The investor can invest in one stock and one risk-free asset. In contrast to most of the existing literature, we assume that the investment opportunity set is not constant and that there are two regimes with

²The bequest can also be interpreted as an exogenous need for cash.

different fundamental parameters such as expected return, volatility, and liquidity. One regime switches to the other regime at the first jump time of a market-independent, but possibly regime-dependent Poisson process.³

We explicitly characterize the solution to the investor’s problem in a general setting.⁴ Our extensive numerical analysis, using parameter estimates from Ang and Bekaert (2002), demonstrates that in contrast to the standard conclusion that transaction costs only have a *second-order* effect, transaction costs may have a *first-order* effect if the investment opportunity set varies stochastically. Specifically, the liquidity premium to transaction cost (LPTC) ratio could be well above 1. The consideration of a time-varying investment opportunity set makes this ratio at least 5 times and in many cases 10 times higher than what Constantinides finds even for the optimal policies.⁵ In addition, we find that the LPTC ratio increases as the bull regime investment opportunity set becomes more attractive than the bear regime investment opportunity set. Intuitively, as the differences in expected return increase, an investor invests more (less) of the stock in the bull (bear) regime, revises the portfolio more dramatically, and thus incurs higher transaction costs. Our analysis thus suggests that consideration of a time varying investment opportunity set is an important ingredient towards understanding higher volume of trade and the relation between transaction costs and the cross section of expected returns. However, our analysis also suggests that the magnitude of liquidity premia cannot be large enough to fully explain the equity premium puzzle even when the investment opportunities are time varying.

We also show that concerns over a potential unpredictable liquidity crash, no matter how unlikely it is, can dramatically reduce investment in stock even when the current market is perfectly liquid. For example, no matter how small the probability of a liquidity

³The investor we consider in this paper can be an institutional investor who does not have any price impact and who updates the estimates of fundamental parameters from time to time.

⁴We also derive closed-form solutions up to some constants in special cases such as the no-intertemporal-consumption case. Results are not reported in this paper to save space, but are available from the authors.

⁵Recall that Constantinides (1986) used a suboptimal consumption policy to emphasize how small the liquidity premium is.

crash is, leverage would become suboptimal no matter how high the equity premium is or how liquid the market is in normal circumstances. Intuitively, an investor who might need to sell her stock position for consumption when the market for the stock is illiquid would incur high transaction costs or even risk insolvency. In addition, we show that if the liquidity crash risk becomes significant, the investor may avoid investing in stock altogether due to the possibility of losing a significant part of her stock investment. This suggests that the existence of liquidity risk may help *partially* explain the Equity-Premium Puzzle and limited participation, which Basak and Cuoco (1998) have shown to be important in explaining the Equity-Premium puzzle. In contrast to the existing literature, our model does not require high risk aversion (e.g., Mehra and Prescott (1985)) or the separation of risk aversion from the intertemporal rate of substitution (e.g., Epstein and Zin (1989)) or habit formation (e.g., Constantinides (1990)) or heterogeneous endowments (e.g., Swan (2005)).⁶

It is worth noting that our above findings suggest that cross regime smoothing is optimal in the presence of transaction costs. Without transaction costs, the optimal investment policy in one regime is independent of parameters in the other regime. In contrast, our analysis indicates that in the presence of transaction costs, an investor optimally responds to changes in one regime by altering investment behavior in both regimes. For example, as the transaction cost in one regime increases (all else equal), the investor trades less often in this regime and trades more in the other regime because the latter regime becomes relatively cheaper to trade in. This finding of cross regime smoothing suggests that the presence of transaction costs can lead to patterns of optimal investment behavior which would seem suboptimal if only the current market conditions were considered.

As far as we know, in the literature with regime-switching, we are the first to provide

⁶Swan (2005) finds that heterogeneous endowments and small transaction costs can help explain the equity premium puzzle after taking into account of the “invisible costs” of foregone trades due to the transaction costs.

a verification theorem for a candidate solution, the steady-state distribution functions for the portfolio holdings, and explicit bounds on the no-transaction regions.

Our theoretical results suggest that extending the two-regime model into a multi-regime model is straightforward but requires significantly more intensive computation. However, the qualitative results obtained in our paper would stay the same. For example, as long as the transaction cost is small relative to the changes in the optimal portfolio target through regimes, we would expect an investor to optimally incur transaction costs when regime switches. This fundamental intuition also applies to the case where the investment opportunity set depends on a continuous state variable. Therefore jumps in the fundamental parameters in the financial market are not critical for our results.

Our paper is related to a large body of literature that addresses the optimal transaction policy for an agent facing transaction costs (see for example, Dumas and Luciano (1991), Schroder (1995), Vayanos (1998), Leland (2000), Lo, Mamaysky, and Wang (2001)). Dumas and Luciano (1991) derive a closed-form solution for an investor with a long term growth objective. Schroder (1995) uses a numerical method to solve for the optimal transaction policy in the presence of fixed costs. Vayanos (1998) derives the asset pricing impact of transaction costs in an overlapping generation framework. Leland (2000) examines a multi-asset investment fund that is subject to transaction costs and capital gains taxes. He develops a relatively simple numerical procedure to compute the no-transaction region. Lo, Mamaysky, and Wang (2001) study the effect of fixed transaction costs on asset prices and find that even small fixed costs can give rise to a significant illiquidity discount on asset prices. All these papers assume that the investment opportunity set is constant. Our analysis suggests that a stochastic investment opportunity set is an important consideration in generating both a higher volume of trade and a greater impact of transaction costs.⁷

⁷Lynch and Balduzzi (2000) examine the impact of stock return predictability and transaction costs on the rebalancing rules, by discretizing time and states to compute the approximate solution. Our analysis complements theirs by providing a transparent and tractable model in which we can explicitly

The rest of the paper is organized as follows. Section 2 presents the model with transaction cost and regime switching and provides characterizations of the solution. Numerical and graphical analysis is presented in Section 3. Section 4 discusses the link between potential illiquidity and the liquidity premium. Section 5 closes the paper. All of the proofs are in the Appendix.

I. Optimal consumption and investment

A. The basic model

Throughout this paper, we assume a probability space $(\Omega, \mathcal{F}, P)$ and a filtration $\{\mathcal{F}_t\}$. Uncertainty in the model is generated by a standard one-dimensional Brownian motion w and by the regime-switch risk. We will assume that w_t and the regime-switch times are adapted.

There are two assets an investor can trade. The first asset (“the bond”) is a money market account, and the second asset (“the stock”) is a risky investment. There are two regimes— “Bull” (regime B) and “bear” (regime b)— and the fundamental parameters in the financial market may be regime-dependent. In addition, we assume that regime i switches into regime j at the first jump time of an independent Poisson process with intensity λ_i , for $i, j \in \{B, b\}$. In regime i , the risk-free interest rate is r_i , and the investor can buy the stock at ask price $S_t^A = (1 + \theta_i)S_t$ and can sell the stock at bid price $S_t^B = (1 - \alpha_i)S_t$, where $\theta_i \geq 0$ and $0 \leq \alpha_i \leq 1$ represent the proportional transaction cost rates, and S_t satisfies

$$\frac{dS_t}{S_t} = \mu_i dt + \sigma_i dw_t, \tag{1}$$

where we assume all parameters are positive constants and $\mu_i > r_i$.

compute bounds on the transaction boundaries, liquidity premia, trading frequency, and expected lifetime transaction cost expenditures.

In regime $i \in \{B, b\}$, when $\theta_i + \alpha_i > 0$, the above model gives rise to equations governing the evolution of the amount invested in the bond, x_t , and the amount invested in the stock, y_t :

$$dx_t = r_i x_t dt - (1 + \theta_i) dI_t + (1 - \alpha_i) dD_t - c_t dt, \quad (2)$$

$$dy_t = \mu_i y_t dt + \sigma_i y_t dw_t + dI_t - dD_t, \quad (3)$$

where the processes D and I represent the cumulative dollar amount of sales and purchases of the stock, respectively. These processes are nondecreasing, right continuous adapted processes with $D(0) = I(0) = 0$. Let x_0 and y_0 be the given initial positions in the bond and the stock respectively. We let $\Theta(x_0, y_0)$ denote the set of admissible trading strategies (c, D, I) such that (2) and (3) are satisfied, $c_t \geq 0$, $\int_0^t c_s ds < \infty$ for all t , and the investor is always solvent, i.e.,⁸

$$x_t + (1 - \alpha_i) y_t \geq 0, \forall t \geq 0 \text{ and } i \in \{B, b\}. \quad (4)$$

Similar to Constantinides (1986), we consider a CRRA investor who derives von Neumann-Morgenstern time additive utility from intertemporal consumption c with weight $1 - k$ and bequest with weight k at death with a time discount rate of ρ . For simplicity, we assume that the mortality rate δ is constant. Then, after integrating out the mortality risk, the investor solves

$$\sup_{(c, D, I) \in \Theta(x_0, y_0)} E \left[\int_0^\infty e^{-(\rho + \delta)t} \left((1 - k) \frac{c_t^{1-\gamma}}{1 - \gamma} + k \delta \frac{(x_t + (1 - \alpha_i) y_t)^{1-\gamma}}{1 - \gamma} \right) dt \right]. \quad (5)$$

⁸The assumption that $\mu_i > r_i$ implies that the investor never shorts the stock and thus that $y_t \geq 0$.

B. Optimal policies with no transaction costs

In this section, we solve the optimal consumption and portfolio selection problem in the absence of transaction costs, i.e., $\theta_i = \alpha_i = 0$, under the regime-switching model presented in the previous section. The results in this section serve as a benchmark for the subsequent analysis.

In this case, the cumulative purchases and sales of the stock can be of infinite variation. Let τ_i be the first jump time since the beginning of regime i . The investor's problem in regime $i \in \{B, b\}$ can be rewritten as

$$V_i(W) = \sup_{\{y_t: t \geq 0\}} E \left[\int_0^{\tau_i} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{W_t^{1-\gamma}}{1-\gamma} \right) dt + e^{-(\rho+\delta)\tau_i} V_j(W_{\tau_i}) \right], \quad (6)$$

subject to

$$dW_t = r_i W_t dt + (\mu_i - r_i) y_t dt + \sigma_i y_t dw_t, \forall i \in \{B, b\}, \quad (7)$$

where $W_t \equiv x_t + y_t \geq 0$, and $V_j(x, y)$ is the value function in regime j .

Under regularity conditions on V_i and V_j , the Hamilton-Jacobi-Bellman (HJB) equations take the form

$$\sup_{(c_i, y_i)} \left\{ \frac{1}{2} \sigma_i^2 y_i^2 V_{iWW} + r_i W V_{iW} + (\mu_i - r_i) y_i V_{iW} - c_i V_{iW} - (\rho + \delta + \lambda_i) V_i + \lambda_i V_j \right. \\ \left. + (1-k) \frac{c_i^{1-\gamma}}{1-\gamma} + k\delta \frac{W_t^{1-\gamma}}{1-\gamma} \right\} = 0, \quad (8)$$

where $i, j \in \{B, b\}$, $i \neq j$. We conjecture

$$V_i(W) = M_i \frac{W^{1-\gamma}}{1-\gamma} \text{ for } M_i > 0, i \in \{B, b\}. \quad (9)$$

By the first-order conditions, we have

$$c_i = \left(\frac{V_{iW}}{1-k} \right)^{-\frac{1}{\gamma}} \quad \text{and} \quad y_i = -\frac{(\mu_i - r_i)V_{iW}}{\sigma_i^2 V_{iWW}}.$$

Then plugging (9) into (8), we obtain that M_i and M_j satisfy the system of equations

$$-(\eta_i + \lambda_i)M_i + \gamma(1-k)^{1/\gamma}M_i^{1-1/\gamma} + \lambda_i M_j + k\delta = 0, \quad (10)$$

where

$$\eta_i = \rho + \delta - (1-\gamma)\left(r_i + \frac{\kappa_i^2}{2\gamma}\right), \quad \kappa_i = \frac{\mu_i - r}{\sigma_i}. \quad (11)$$

To ensure the existence of an optimal solution, we adopt the following assumption, similar to that of Merton (1971).

Assumption 1. $\eta_i > 0, \forall i \in \{B, b\}$.

Lemma 1. *Under Assumption 1, there is a unique solution (M_B, M_b) to (10).*

Proof: see Appendix.

The following verification theorem shows that indeed our conjecture is correct.

Theorem 1. *Under Assumption 1 for regime $i \in \{B, b\}$, V_i defined in (9) is the value function defined in (6). In addition, the optimal consumption policy is $c_i^* = \left(\frac{M_i}{1-k}\right)^{-\frac{1}{\gamma}} W$ and the optimal fraction of wealth invested in the stock is $\pi_i^* = \frac{\mu_i - r_i}{\gamma\sigma_i^2}$.*

Proof: This follows from an argument similar to (although simpler than) that presented in the proof of Theorem 2..

Theorem 1. implies that both the optimal consumption and the optimal dollar amount invested in the stock are constant fractions of the investor's wealth in each regime. Even though the investor smooths consumption across regimes (as reflected in that M_i and M_j are jointly determined by equation (10)), the optimal investment policy is myopic in the sense that π_i^* only depends on the current regime parameters. This follows from the fact that the risk of regime switching is unhedgeable using the existing securities and that the investor can rebalance at regime switching time without any transaction costs. As will be shown later, the presence of transaction costs makes the optimal trading strategy no longer myopic.

C. Optimal policies with transaction costs

Now suppose there are transaction costs in both regimes, i.e., $\theta_i > 0, \alpha_i > 0, i \in \{B, b\}$. Using similar notations to those in the previous section, we can rewrite the investor's problem as

$$v_i(x, y) = \sup_{(c, D, I)} E \left[\int_0^{\tau_i} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt + e^{-(\rho+\delta)\tau_i} v_j(x_{\tau_i}, y_{\tau_i}) \right], \quad (12)$$

subject to (2)-(4), where $v_i(x, y)$ denotes the value function in regime i .

It can be easily verified that the value functions v_B and v_b are concave and homogeneous of degree $1 - \gamma$ in (x, y) (cf. Fleming and Soner (1993), Lemma VIII.3.2). As in Davis and Norman (1990) and Liu and Loewenstein (2002), the solvency region $\mathcal{S}_i$ splits into three regions: a 'no-trading' (NT_i) region, a 'buy' (B_i) region, and a 'sell' (S_i) region. The homogeneity of v_i implies that the transaction boundaries are straight lines (see Figure 1). In addition, there exists an interval $[\underline{z}_i, \bar{z}_i]$ such that in regime i , the investor trades only

the minimum amount to keep the bond to stock ratio

$$z \equiv \frac{x}{y}$$

inside the interval. We depict this analysis in Figures 2 and 3. Figure 2 shows the solvency region when the two no-transaction regions (NT_B and NT_b) are separated. Intuitively, this case occurs when the difference between the two regimes is large, the regime transition probability is low (e.g, the expected return on stock in the Bull regime is sufficiently larger than that in the bear regime), and the transaction cost is relatively small. As the transition probability increases, the no-transaction regions move toward each other and eventually become overlapped, as shown in Figure 3.

The above optimal transaction policy implies that the HJB equation takes the following form

$$\frac{1}{2}\sigma_i^2 y^2 v_{iyy} + r_i x v_{ix} + \mu_i y v_{iy} + \frac{\gamma(1-k)^{1/\gamma}}{1-\gamma} v_{ix}^{1-1/\gamma} - (\rho + \delta) v_i + k \delta \frac{(x + (1 - \alpha_i)y)^{1-\gamma}}{1-\gamma} + \lambda_i (v_j - v_i) = 0, \quad (13)$$

for $j \neq i$ in NT_i . In the sell region, the investor transacts immediately to the sell boundary. Therefore,

$$v_i(x, y) = A_i \frac{(x + (1 - \alpha_i)y)^{1-\gamma}}{1-\gamma}, \quad (14)$$

where A_i is a positive constant to be determined. Similarly, in the buy region, the investor transacts immediately to the buy boundary. Therefore,

$$v_i(x, y) = B_i \frac{(x + (1 + \theta_i)y)^{1-\gamma}}{1-\gamma}, \quad (15)$$

where B_i is also a positive constant to be determined.

By the homogeneity of the value functions, there exists a function $\psi_i : (\underline{z}_i, \bar{z}_i) \rightarrow \mathbb{R}$ in

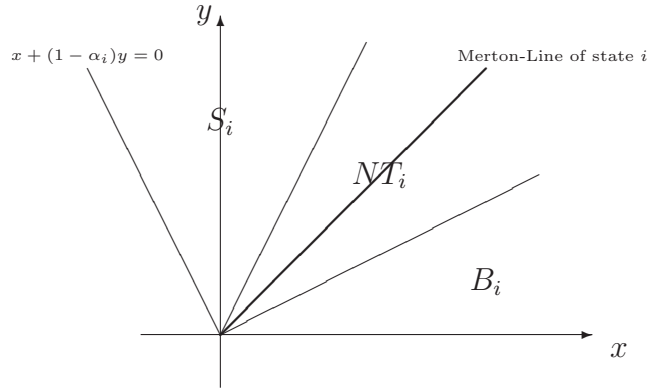


Figure 1: The solvency region splits into the regions B_i , S_i , NT_i

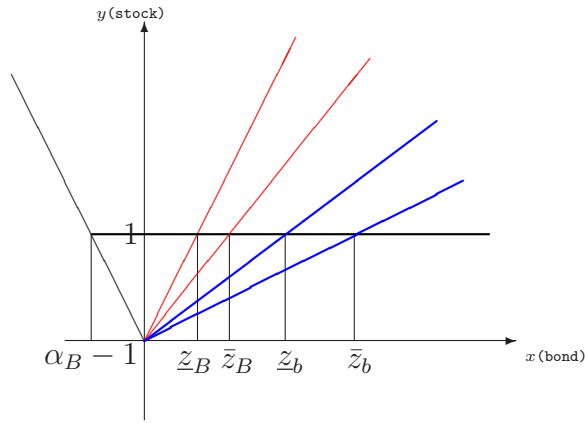


Figure 2: The solvency region of a separated case when $\alpha_B = \alpha_b$

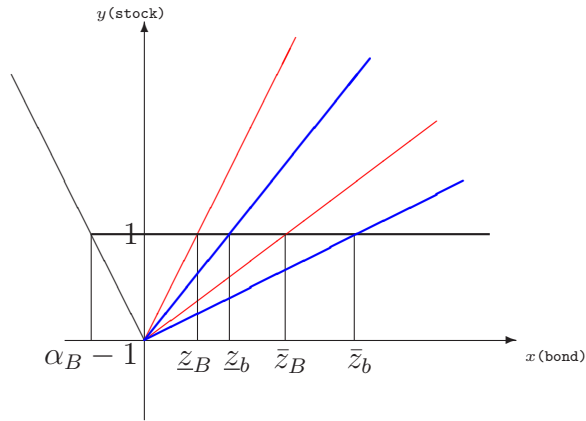


Figure 3: The solvency region of an overlapped case when $\alpha_B = \alpha_b$

regime i satisfying

$$v_i(x, y) \equiv y^{1-\gamma} \psi_i\left(\frac{x}{y}\right). \quad (16)$$

This implies that

$$\psi_i(z) = \begin{cases} A_i \frac{(z+(1-\alpha_i))^{1-\gamma}}{1-\gamma} & z < \underline{z}_i \\ B_i \frac{(z+(1+\theta_i))^{1-\gamma}}{1-\gamma} & z > \bar{z}_i. \end{cases} \quad (17)$$

Using Equation (16) and the ratio z , we can simplify the partial differential equation (PDE) to get the following ordinary differential equation (ODE) in NT_i :

$$\beta_2^i z^2 \psi_i''(z) + \beta_1^i z \psi_i'(z) + \beta_0^i \psi_i(z) + \frac{\gamma(1-k)^{1/\gamma}}{1-\gamma} \psi_i'(z)^{1-1/\gamma} + k\delta \frac{(z+(1-\alpha_i))^{1-\gamma}}{1-\gamma} + \lambda_i \psi_j(z) = 0, \quad (18)$$

$i \neq j$ for $z \in (\underline{z}_i, \bar{z}_i)$, where $\beta_2^i = \frac{1}{2}\sigma_i^2$, $\beta_1^i = \gamma\sigma_i^2 - (\mu_i - r_i)$, $\beta_0^i = \frac{1}{2}\sigma_i^2\gamma(\gamma-1) + (1-\gamma)\mu_i - \rho - \delta - \lambda_i$. Then we have the following verification theorem.⁹

Theorem 2. *For $i, j \in \{B, b\}$ and $j \neq i$, suppose we have concave, increasing, and homothetic $C^{2,2}$ solutions to*

$$\frac{1}{2}\sigma_i^2 y^2 v_{iyy} + r_i x v_{ix} + \mu_i y v_{iy} + \frac{\gamma(1-k)^{1/\gamma}}{1-\gamma} v_{ix}^{1-\frac{1}{\gamma}} - (\rho+\delta)v_i + k\delta \frac{(x+(1-\alpha_i)y)^{1-\gamma}}{1-\gamma} + \lambda_i(v_j - v_i) \leq 0 \quad (19)$$

with equality for $\frac{x}{y} \in [\underline{z}_i, \bar{z}_i]$, which satisfy

$$(1+\theta_i)v_{ix} \geq v_{iy} \quad (20)$$

with equality for $\frac{x}{y} > \bar{z}_i$, and

$$(1-\alpha_i)v_{ix} \leq v_{iy} \quad (21)$$

⁹When $k = 1$, a fairly explicit solution can be computed.

with equality for $\frac{x}{y} < \underline{z}_i$. Then v_i is the value function; the optimal consumption is given by

$$c^* = \left(\frac{v_{ix}}{1-k} \right)^{-\frac{1}{\gamma}}; \quad (22)$$

and the optimal transaction policy is to transact the minimum amount so as to keep $\frac{x}{y}$ between $\underline{z}_i$ and $\bar{z}_i$.

Proof: See Appendix.

Since the value functions ψ_B and ψ_b need to be solved simultaneously through the coupled ODEs for the two regimes, the value function and thus the optimal trading strategy in one regime are both affected by the investment opportunity set in the *other* regime. Therefore, in the presence of transaction costs, the optimal trading strategy is no longer myopic, and the investor smooths not only consumption but also trading strategy across regimes.

The following proposition provides some bounds on the value functions and on the optimal no-transaction boundaries for the leading cases we analyze later. Similar bounds can be developed for a more general specification of our model. These bounds facilitate numerical computation by suggesting useful starting values for the boundaries. They also provide useful information on the width of the no-transaction regions. In addition, the proof suggests that similar bounds can be obtained using the same approach when there are more than two regimes.

Proposition 1. *Suppose $\gamma > 1$, $k = \delta = 0$, $r_B = r_b = r$, $\theta_B = \theta_b = \theta$, and $\alpha_B = \alpha_b = \alpha$. Then assuming we have a solution satisfying the conditions of Theorem 2., we have the following bounds.*

1. For any $1 - \alpha \leq \zeta \leq 1 + \theta$ and $i \in \{b, B\}$,

$$\frac{C}{1 - \gamma}(x + (1 - \alpha)y)^{1 - \gamma} \leq v_i(x, y) \leq \frac{M_i}{1 - \gamma}(x + \zeta y)^{1 - \gamma}, \quad (23)$$

where $C = (\frac{\rho - (1 - \gamma)r}{\gamma})^{-\gamma}$ and M_i is given in Lemma 1..

2. If $\sigma_i^2 < \sigma_j^2$ and $\mu_i \geq \mu_j$, then $v_i(x, y) \geq v_j(x, y)$ and

$$\underline{z}_i \geq (1 - \alpha) \left(\frac{\gamma \sigma_i^2}{2(\mu_i - r)} - 1 \right). \quad (24)$$

3. For $i, j \in \{b, B\}$, $i \neq j$. If $\underline{z}_i \leq \underline{z}_j$, then

$$\underline{z}_i \leq (1 - \alpha) \left(\frac{\gamma \sigma_i^2}{\mu_i - r} - 1 \right). \quad (25)$$

4. For $i, j \in \{b, B\}$, $i \neq j$, if $\bar{z}_i \geq \bar{z}_j$, then

$$\bar{z}_i \geq (1 + \theta) \left(\frac{\gamma \sigma_i^2}{\mu_i - r} - 1 \right). \quad (26)$$

Proof: See Appendix.

II. Liquidity premium, trading strategy, and transaction costs

Constantinides (1986) introduces the concept of liquidity premia to measure the effect of transaction costs on expected returns. He defines the liquidity premium to be the maximum expected return an investor is willing to exchange for zero transaction cost. He concludes that transaction costs have only a second-order effect on investors' utility, i.e., the

ratio of the liquidity premium to the transaction cost (LPTC) rate is much smaller than 1. However, he obtains this result under the assumption of a constant investment opportunity set, which tends to decrease the effect of transaction cost because the investor trades infrequently. In contrast to that of Constantinides (1986), the investment opportunity set in our model changes stochastically. This time-varying opportunity set may induce an investor to trade more frequently or alter more significantly her trading strategy and thus may produce a first-order effect of the transaction cost.

Following Constantinides (1986), we define the liquidity premium to be the maximum expected return that an investor is willing to give up in both regimes to get rid of the transaction costs.

Definition 1 *Let $\Delta_i(x_0, y_0)$ be the liquidity premium in regime i at (x_0, y_0) for $i \in \{B, b\}$. Then Δ_i is such that*

$$v_i(x_0, y_0) = M_i \frac{(x_0 + y_0)^{1-\gamma}}{1-\gamma}, \quad (27)$$

where M_i is the unique solution of (10) with

$$\kappa_i = \frac{(\mu_i - \Delta_i - r_i)^+}{\sigma_i}, \kappa_j = \frac{(\mu_j - \Delta_i - r_j)^+}{\sigma_j}, \quad (28)$$

and $j \neq i$.

In this measure, we take the positive parts because an investor's utility is minimized when the risk premium is zero.

By the homogeneity of the value function, the liquidity premium only depends on the initial fraction $z = \frac{x_0}{y_0}$. We use the steady-state distribution of z_t to compute the average liquidity premium.¹⁰ This can be interpreted as a cross-sectional liquidity premium average for different investors. Let $\Delta_i(x_0, y_0)$ be the liquidity premium in regime i at (x_0, y_0) for

¹⁰We have also computed liquidity premia at the Merton line and on $y_0 = 0$ as Constantinides (1986) and obtained similar results.

$i \in \{B, b\}$, as defined in Definition 1. Then the average liquidity premium $\bar{\Delta}_i$ in regime i for $i \in \{B, b\}$ is

$$\bar{\Delta}_i = \int_{\underline{z}_i}^{\bar{z}_i} \Delta_i(z, 1) \frac{\lambda_B + \lambda_b}{\lambda_j} \phi_i(z) dz, \quad (29)$$

where $j \neq i$ and $\phi_i(z)$ is the steady state density of z in regime i , which is characterized in Proposition 4. in the Appendix. The average liquidity premium across both regimes is

$$\bar{\Delta} = \frac{\lambda_b}{\lambda_B + \lambda_b} \bar{\Delta}_B + \frac{\lambda_B}{\lambda_B + \lambda_b} \bar{\Delta}_b. \quad (30)$$

A. Optimal Policies in the Benchmark Case

Our analysis uses the parameters similar to those estimated for the US equity market in Ang and Bekaert (2002) as the benchmark case: $\mu_B = 0.1394$, $\mu_b = 0.1394$, $\sigma_B = 0.1313$, $\sigma_b = 0.2600$, $\lambda_B = 0.2353$, $\lambda_b = 1.7391$, $\rho = 0.1$, $\gamma = 2.0$, $r_B = r_b = 0.05$, $\theta_B = \alpha_B = 0.01$, $\theta_b = \alpha_b = 0.01$, $k = 0$, and $\delta = 0$. These parameters reflect both that volatility is significantly higher in the bear market regime than in the Bull market regime and that the null hypothesis that the mean returns are the same across regimes cannot be rejected by data.¹¹ As in Constantinides (1986), we set $k = 0$ and $\delta = 0$ so that the utility is only from consumption and the investor has an infinite horizon.¹² The Poisson jump intensities are chosen to be consistent with the typical pattern that a Bull regime lasts longer (on average 4.25 years) than a bear regime (on average 0.58 years). In our calibrated model, an investor can infer about the state of the current regime from observing stock return volatility, which is the only parameter that varies across regimes in our benchmark case.

In our benchmark case, it is straightforward to calculate the optimal policies with no transaction costs by solving the system of equations given by (10). The optimal portfolio

¹¹To address any concerns that the estimation for the expected return μ_b in the bear regime might be imprecise, we vary μ_b in Figure 9.

¹²It is well known that the investor's horizon can affect liquidity premia (see Liu and Loewenstein (2002) for example). We focus on an infinite horizon case to approximate the case with a long horizon.

policies are myopic: the investor optimally maintains a ratio of stock investment to wealth of 2.5923 in the Bull regime and of 0.6612 in the bear regime. The optimal consumption across regimes displays cross-regime smoothing: the consumption-to-wealth ratio is 12.793% in the Bull market and 12.530% in the bear market.¹³

For our benchmark case with transaction costs, we use an iterative method to solve the coupled free boundary value problem described in equations (17) and (18). Table 1 depicts the optimal policy for the case with transaction costs as well as how this policy changes with different parameters. With transaction costs, the optimal consumption to liquidated wealth ratios are around 12.5%. For our benchmark case, we also see cross-regime smoothing in the portfolio holdings. The investor optimally maintains the ratio of the dollar amount invested in the stock to liquidated wealth ($\frac{y}{x+(1-\alpha_B)y}$) between 1.9469 and 2.8934 in the Bull regime and between 0.6005 and 0.9183 in the bear regime. Compared with the optimal policy in the bear regime with no transaction costs, the optimal policy tends to hold more stock in the bear regime both because of the costly rebalancing when there are transaction costs and because the bear regime is expected to last a relatively short time. This cross-regime smoothing is clearly indicated in Figure 4, which plots the behavior of the bear transaction boundaries against changes in transaction costs in the Bull market. There is a clear sensitivity to the Bull market transaction costs in the bear market optimal portfolio policy. In fact, when the Bull market transaction cost is relatively high compared to the bear market transaction cost, the optimal transaction boundary in the bear market is entirely above the myopic policy (the “Merton Line”) in the no-transaction-cost case. This possibility arises because it is cheaper to buy the asset in the bear regime, which is expected to last a short time, and enjoy the benefits in the Bull regime.

¹³In a model with no regime switching, the optimal consumption-to-wealth ratio for parameters equal to those in the Bull regime is 13.295%, and the optimal consumption-to-wealth ratio for parameters equal to those in the bear regime is 8.978% .

Parameters	c_B^*/W_B at		c_b^*/W_b at		y_B^*/W_B at		y_b^*/W_b at		Average LPTC Ratio	
	$\bar{z}_B$	$\bar{z}_B$	$\bar{z}_b$	$\bar{z}_b$	$\bar{z}_B$	$\bar{z}_B$	$\bar{z}_b$	$\bar{z}_b$	$\bar{\Delta}_B/\alpha_B$	$\bar{\Delta}_b/\alpha_b$
Benchmark	0.1244	0.1264	0.1204	0.1210	2.8934	1.9469	0.9183	0.6005	0.4399	0.5040
μ_B										
×1.05	0.1320	0.1343	0.1271	0.1276	3.1584	2.0923	0.9163	0.5981	0.4824	0.5531
×0.95	0.1174	0.1191	0.1143	0.1148	2.6312	1.7986	0.9202	0.6027	0.3980	0.4555
σ_B										
×1.05	0.1203	0.1221	0.1169	0.1175	2.5905	1.7847	0.9194	0.6018	0.4046	0.4634
×0.95	0.1292	0.1314	0.1245	0.1250	3.2489	2.1321	0.9171	0.5990	0.4771	0.5467
σ_b										
×1.03	0.1242	0.1262	0.1202	0.1207	2.8936	1.9470	0.8690	0.5623	0.4491	0.5155
×0.97	0.1246	0.1266	0.1208	0.1214	2.8932	1.9467	0.9738	0.6423	0.4293	0.4909
θ_b										
×1.05	0.1244	0.1264	0.1204	0.1210	2.8934	1.9469	0.9183	0.5942	0.4392	0.5034
×0.95	0.1244	0.1264	0.1204	0.1210	2.8934	1.9469	0.9183	0.6068	0.4399	0.5040
λ_b										
×1.05	0.1246	0.1266	0.1207	0.1213	2.8932	1.9467	0.9304	0.6027	0.4386	0.5029
×0.95	0.1242	0.1262	0.1201	0.1207	2.8936	1.9471	0.9063	0.5981	0.4405	0.5043
γ										
×1.05	0.1233	0.1251	0.1194	0.1198	2.7292	1.8664	0.8768	0.5697	0.4194	0.4836
×0.95	0.1256	0.1278	0.1216	0.1222	3.0774	2.0342	0.9651	0.6347	0.4429	0.5071
ρ										
×1.05	0.1270	0.1290	0.1229	0.1235	2.8905	1.9437	0.9183	0.5998	0.4399	0.5051
×0.95	0.1219	0.1238	0.1180	0.1185	2.8964	1.9501	0.9183	0.6013	0.4392	0.5022

Table 1: The benchmark case is with parameter $r_B = r_b = 0.05$, $\mu_B = \mu_b = 0.1394$, $\sigma_B = 0.1313$, $\sigma_b = 0.2600$, $\gamma = 2.0$, $\lambda_B = 0.2353$, $\lambda_b = 1.7391$, $\rho = 0.1$, $k = 0$, $\delta = 0$, and $\theta_B = \alpha_B = \theta_b = \alpha_b = 0.01$. Here, $W_i = x + (1 - \alpha_i)y$ is the value of liquidated wealth, $\alpha = (\lambda_b \alpha_B + \lambda_B \alpha_b)/(\lambda_B + \lambda_b)$, $\bar{\Delta}_i$ represents the average liquidity premium for the whole inaction interval of regime $i \in \{B, b\}$, and $\bar{\Delta}$ is the average liquidity premium across both regimes.

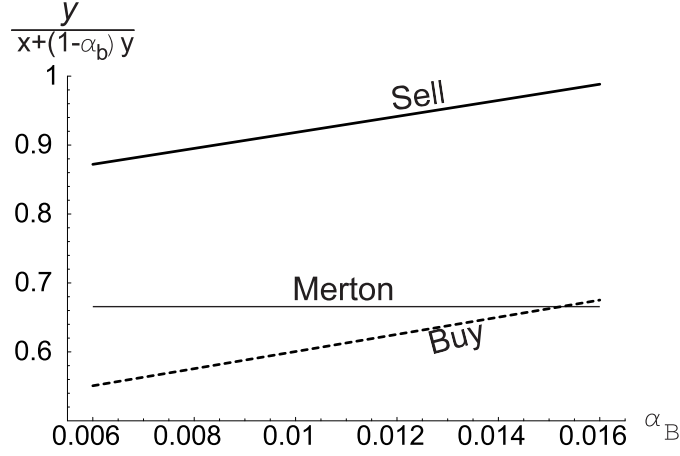


Figure 4: The fraction of wealth invested in the stock in the bear regime, as a function of the transaction cost in the Bull regime for parameters: $\mu_B = 0.1394$, $\mu_b = 0.1394$, $\sigma_B = 0.1313$, $\sigma_b = 0.2600$, $\lambda_B = 0.2353$, $\lambda_b = 1.7391$, $\rho = 0.1$, $\gamma = 2.0$, $r_B = r_b = 0.05$, $\theta_B = \alpha_B$, $\theta_b = \alpha_b = 0.01$, $k = 0$, and $\delta = 0$.

Table 1 provides further information on how the optimal policies change with model parameters. While the general comparative statics follow intuitively, changes in the Bull regime parameters have a relatively larger impact on the optimal policies than those in the bear regime because the Bull regime is expected to last longer.

Table 1 also displays the liquidity premia averaged across the steady state distribution of portfolio holdings. Regime switching increases the liquidity premia 4-5 times those reported in Constantinides (1986). Our following sections examine this finding in more detail.

B. Liquidity premia in the regime switch model

Figure 5 plots the steady-state average liquidity premia against the transaction cost in the Bull regime. This figure shows that when the transaction cost is small, the liquidity premium to transaction cost ratio is four to six times higher than that found in Constan-

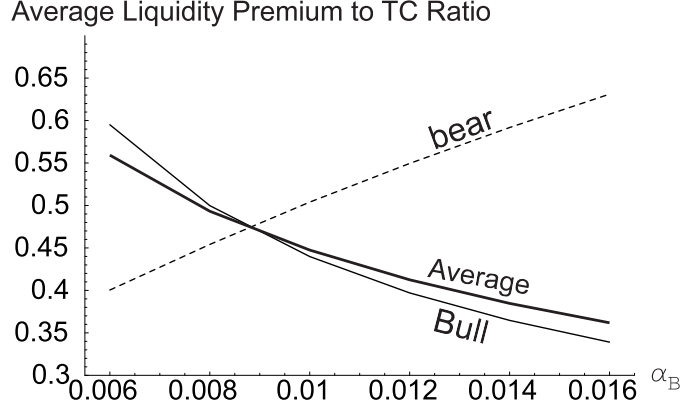


Figure 5: The liquidity premium to transaction cost ratio, as a function of the transaction cost in the Bull regime for parameters: $\mu_B = 0.1394$, $\mu_b = 0.1394$, $\sigma_B = 0.1313$, $\sigma_b = 0.2600$, $\lambda_B = 0.2353$, $\lambda_b = 1.7391$, $\rho = 0.1$, $\gamma = 2.0$, $r_B = r_b = 0.05$, $\theta_B = \alpha_B$, $\theta_b = \alpha_b = 0.01$, $k = 0$, and $\delta = 0$.

tinides (1986). This suggests that a time-varying investment opportunity set is a much more important determinant of the liquidity premium than previously thought.

The fundamental intuition that leads to the small liquidity premium in Constantinides (1986) is that an investor optimally reduces trading frequency and thus reduces transaction cost payment. Figure 6 shows the expected time from purchase to sale (see Appendix for the derivation of this measure) as a function of risk aversion in the Constantinides model and in our model. In this figure, the Constantinides model corresponds to a model with no regime shifts and its volatility is equal to the unconditional volatility across regimes. When the investment opportunity set is not constant, the expected holding period for stock is significantly shorter, which is consistent with the higher liquidity premium found in Figure 5.

To confirm that the changing opportunity set leads to higher transaction costs, Figure 7 displays the expected discounted lifetime transaction costs as a fraction of the liquidated wealth (see Appendix for the derivation of this measure) against risk aversion in our model

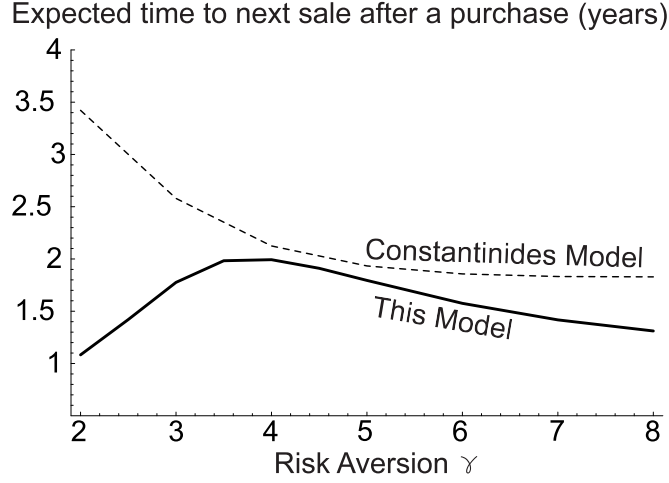


Figure 6: The expected time from purchase to sale, as a function of the risk aversion for parameters: $\mu_B = 0.1394$, $\mu_b = 0.1394$, $\sigma_B = 0.1313$, $\sigma_b = 0.2600$, $\lambda_B = 0.2353$, $\lambda_b = 1.7391$, $\rho = 0.1$, $r_B = r_b = 0.05$, $\theta_B = \alpha_B = 0.01$, $\theta_b = \alpha_b = 0.01$, $k = 0$, and $\delta = 0$. The Constantinides model corresponds to the one-regime case with $\sigma = 0.1466$.

and in the Constantinides model. Using a continuously compounded discount rate of 12%, we see that for low risk aversion levels, the expected discounted lifetime transaction costs can amount to 4% of initial wealth versus only about 0.5% in the unconditional model without regime shifts.

One might wonder how large the liquidity premia can get in our model. To get very large liquidity premia, our previous analysis suggests that we need a fairly large difference between the investment opportunity sets, a relatively low transaction cost, and relatively frequent transitions between regimes. Figure 8 shows how the liquidity premium to transaction cost ratio changes as we vary the frequency of switching from the Bull regime to the bear regime. It suggests that the changing opportunity set alone cannot generate a liquidity premium much greater than 1.6% with reasonable parameters. To generate a 1.6% liquidity premium, one would need to assume that the Bull regime lasts only about 0.6 years, which is much shorter than the time frame empirically observed. Nevertheless, it also

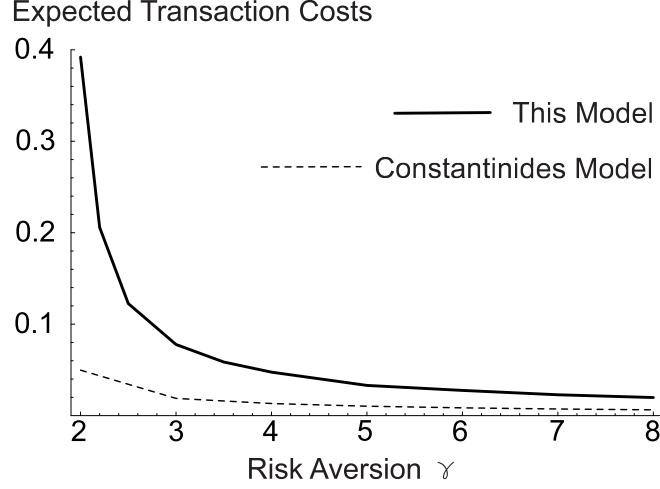


Figure 7: The expected discounted transaction costs as the fraction of the liquidated wealth against the risk aversion for parameters: $\mu_B = 0.1394$, $\mu_b = 0.1394$, $\sigma_B = 0.1313$, $\sigma_b = 0.2600$, $\lambda_B = 0.2353$, $\lambda_b = 1.7391$, $\rho = 0.1$, $r_B = r_b = 0.05$, $\theta_B = \alpha_B = 0.01$, $\theta_b = \alpha_b = 0.01$, $k = 0$, $\delta = 0$, and $\nu = 0.12$. The Constantinides model corresponds to the one-regime case with $\sigma = 0.1466$.

shows that transaction costs, along with a stochastically varying investment opportunity set, can play a more important role than previously thought.

Figure 9 plots the steady-state average liquidity premia against the expected return in the bear regime. This figure shows that when the expected return in the bear regime decreases, the liquidity premium becomes even greater. This is because as the investment opportunity sets differ more across regimes, the no-transaction intervals become more separated, and an investor incurs higher transaction costs at the regime switching time.

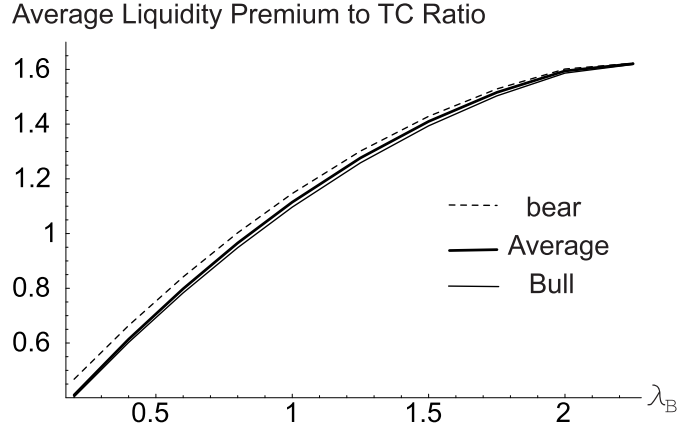


Figure 8: The liquidity premium to transaction cost ratio, as a function of the Bull regime switch intensity for parameters: $\mu_B = 0.1394$, $\mu_b = 0.1394$, $\sigma_B = 0.1313$, $\sigma_b = 0.2600$, $\lambda_b = 1.7391$, $\rho = 0.1$, $\gamma = 2.0$, $r_B = r_b = 0.05$, $\theta_B = \alpha_B = 0.01$, $\theta_b = \alpha_b = 0.01$, $k = 0$, and $\delta = 0$.

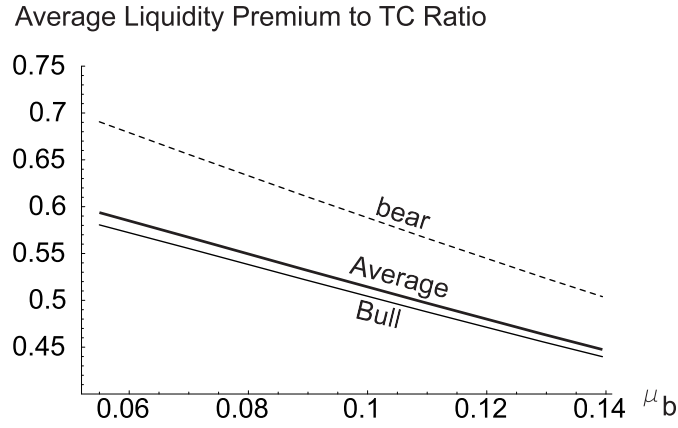


Figure 9: The liquidity premium to transaction cost ratio, as a function of the expected return in the bear regime for parameters: $\mu_B = 0.1394$, $\sigma_B = 0.1313$, $\sigma_b = 0.2600$, $\lambda_B = 0.2353$, $\lambda_b = 1.7391$, $\rho = 0.1$, $\gamma = 2.0$, $r_B = r_b = 0.05$, $\theta_B = \alpha_B = 0.01$, $\theta_b = \alpha_b = 0.01$, $k = 0$, and $\delta = 0$.

III. Liquidity risk, limited participation, and equity-premium puzzle

There is a large literature on the Equity-Premium Puzzle (Risk-free Rate Puzzle) first identified by Mehra and Prescott (1985), including those using separation of the risk aversion from the intertemporal rate of substitution (e.g., Epstein and Zin (1989)) and those using habit formation (e.g., Constantinides (1990)). As shown below, our model suggests that the existence of liquidity risk may help partially explain the puzzle. In essence, a high liquidity premium may arise because the optimal portfolio composition may be dramatically affected by liquidity risk.

To clarify the basic intuition that our model can potentially help explain the puzzle, suppose there are two liquidity regimes where liquidity changes across the regimes. Suppose further that in the “Liquid” regime (in a slight abuse of our previous notation $i = L$, $\alpha_L = \theta_L = 0$), the market is perfectly liquid (i.e., no transaction cost), and in the “Illiquid” regime, the market is perfectly illiquid ($i = I$, $\alpha_I = 1$, and $\theta_I = \infty$). At any point in time, there is a positive (but maybe very small) probability of switching from the Liquid regime to the Illiquid regime (i.e., the Liquid regime Poisson intensity $\lambda_L > 0$) and a zero probability of switching back ($\lambda_I = 0$). In addition, to isolate the effect of liquidity risk, we assume that only liquidity and expected duration change across regimes (i.e., $\mu_L = \mu_I = \mu$, $\sigma_L = \sigma_I = \sigma$, and $r_L = r_I = r$). Then we have the following result.

Proposition 2. *Suppose $k = 0$, $\alpha_I = 1$, $\theta_I = \infty$, $\alpha_L = \theta_L = 0$, $\lambda_L > 0$, and $\lambda_I = 0$. Then the value function in the illiquid regime is*

$$v_I(x, y) = \left(\frac{\gamma}{\rho - (1 - \gamma)r} \right)^\gamma \frac{x^{1-\gamma}}{1 - \gamma}. \quad (31)$$

In addition,

(i) if $\lambda_L \geq \mu - r$, then the value function in the liquid regime is

$$v_L(x, y) = \left(\frac{\gamma}{\rho - (1 - \gamma)r} \right)^\gamma \frac{(x + y)^{1-\gamma}}{1 - \gamma}, \quad (32)$$

and it is optimal not to invest any amount in the stock, i.e.,

$$z_L^* = \infty; \quad (33)$$

(ii) if $\lambda_L < \mu - r$, then the value function in the liquid regime is

$$v_L(x, y) = A_L \frac{(x + y)^{1-\gamma}}{1 - \gamma}, \quad (34)$$

and the optimal bond-to-stock ratio is $z_L^* > 0$, where A_L and z_L^* solve (using the same definitions of β_j^L as in equation (18))

$$-A_L \gamma \beta_2^L z_L^{*2} (z_L^* + 1)^{-\gamma-1} + A_L \beta_1^L z_L^* (z_L^* + 1)^{-\gamma} + (A_L \beta_0^L + \gamma A_L^{1-1/\gamma}) \frac{(z_L^* + 1)^{1-\gamma}}{1 - \gamma} + \lambda_L \psi_I(z_L^*) = 0. \quad (35)$$

and

$$A_L((\mu_L - r_L)(z_L^* + 1) - \gamma \sigma_L^2) = \lambda_L (z_L^* + 1)^{\gamma+2} \psi_I'(z_L^*) - \lambda_L (1 - \gamma)(z_L^* + 1)^{\gamma+1} \psi_I(z_L^*). \quad (36)$$

Proof: See Appendix

This proposition suggests that if the probability of a liquidity crash is large enough, then it may be optimal for the investor *not* to invest in the stock at all, even in the perfectly liquid regime with a high expected excess return. This implies that the widely documented limited participation in the stock market may be due to concerns over liquidity crash, and

the high observed equity premium may be caused by the high liquidity risk premium. Therefore, our model can potentially help explain the Equity Premium puzzle and the limited participation anomaly.

Similar intuition implies that even if the liquidity shock was not as extreme as above, the investor would still reduce significantly her stock investment for the fear of the liquidity shock. As a result, concerns over a potential liquidity crash, no matter how unlikely it is, can dramatically reduce investment in stock, even when the current market is perfectly liquid and the expected excess return is high.

The following proposition shows that the possibility of switching to the Illiquid regime will rule out leverage in the Liquid regime as long as $\lambda_L > 0$ and λ_I is finite. In other words, as long as there is a strictly positive probability, no matter how small it is, of switching to the Illiquid regime and staying there for a positive amount of time, the investor will never lever up in the perfectly liquid regime, no matter how high the risk premium is. This result in turn leads to a lower bound on the liquidity premium in the case where leverage would be optimal in a model without liquidity crashes.

Proposition 3. *Suppose $k = 0$, $\lambda_L > 0$, and $\lambda_I < \infty$. Then there cannot be any leverage in the liquid regime no matter how large the risk premium is. In addition, if $\mu - r > \gamma\sigma^2$, then the liquidity premium in the liquid regime is greater than $\mu - r - \sigma\sqrt{2\gamma(\mu - r - \frac{\gamma\sigma^2}{2})} > 0$.*

It is worth noting that the lower bound applies regardless of the duration of the Illiquid regime (as long as it lasts for a positive amount of time). In addition, the lower bound can also be significant. For example, in a model with $\mu - r = 0.1$, $\sigma = 0.15$, and $\gamma = 2$, the lower bound on the liquidity premium is 1.65%, even if the probability of switching to the illiquid regime is very small and if the expected duration of the illiquid regime is very short.

To isolate the impact of time-varying liquidity, we use for our benchmark case the model whose volatility of stock returns equal the unconditional value of $\sigma_L = \sigma_I = 0.1466$ and $\mu_L = \mu_I = 0.1394$. The average duration of the perfectly liquid regime is $\frac{1}{\lambda_L} = 2$ years. In this case, the lower bound on the liquidity premium given by Proposition 3. is 1.3%. Figure 10 plots the fraction of wealth in stock as a function of the Poisson regime switching intensity λ_I in the Illiquid regime. This figure shows that even though in the absence of a liquidity crash the investor would lever up to invest 208% of her wealth in the stock, with the potential liquidity crash, the investor never lever up, even when the duration of the Illiquid regime is short (i.e., λ_I is high).¹⁴

Figure 11 plots the average liquidity premium $\bar{\Delta}_L$ as a function of the Poisson regime switching intensity λ_I in the Illiquid regime. This figure shows that the investor is willing to pay a significant portion of the risk premium to avoid an unlikely liquidity crash. For example, out of the 8.94% risk premium, the investor is willing to forgo more than 2% to avoid the liquidity crash when the expected duration of the Illiquid regime is 2 weeks (i.e., $\lambda_I = 30$), even though such a crash occurs only once in every 2 years ($\lambda_B = 0.5$) on average.

IV. Conclusion

Standard portfolio selection literature (e.g., Constantinides (1986)) finds that although transaction costs alter trading strategies significantly, they only have a second-order effect on the liquidity premia. In this paper, we show that this conclusion depends crucially on the assumption of a constant investment opportunity set. In contrast, in a stochastic regime-switching model with transaction costs, we show that transaction costs can have a *first-order* effect on liquidity premia for a wide range of parameters. This suggests that

¹⁴The regime switching intensity value λ_L of 0.5 means that on average the investor is going to stay in the perfectly liquid regime for 2 years before a liquidity crash occurs.

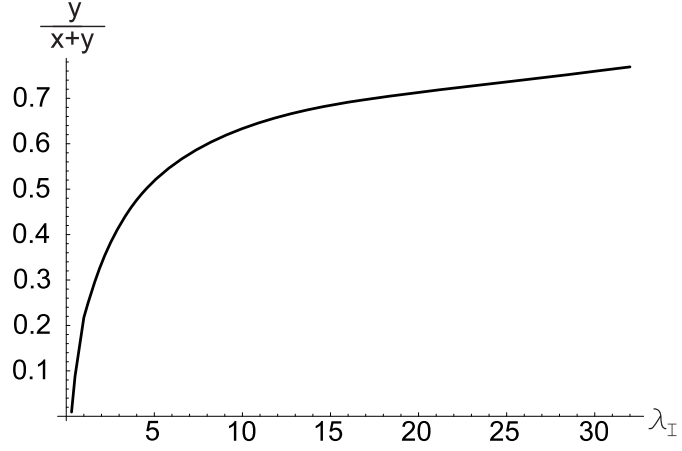


Figure 10: The fraction of wealth in stock, as a function of the Poisson regime switching intensity in the illiquid regime λ_I for parameters: $\mu_L = \mu_I = 0.1394$, $\sigma_L = \sigma_I = 0.1466$, $\lambda_L = 0.5$, $\rho = 0.1$, $\gamma = 2.0$, $r_B = r_b = 0.05$, $\theta_L = \alpha_L = 0$, $\theta_I = \infty$, $\alpha_I = 1$, $k = 0$, and $\delta = 0$.

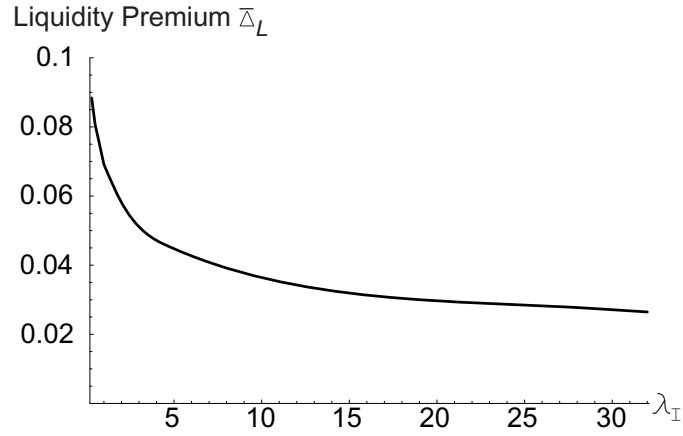


Figure 11: The liquidity premium $\bar{\Delta}_L$, as a function of the Poisson regime switching intensity in the illiquid regime λ_I for parameters: $\mu_L = \mu_I = 0.1394$, $\sigma_L = \sigma_I = 0.1466$, $\lambda_L = 0.5$, $\rho = 0.1$, $\gamma = 2.0$, $r_B = r_b = 0.05$, $\theta_L = \alpha_L = 0$, $\theta_I = \infty$, $\alpha_I = 1$, $k = 0$, and $\delta = 0$.

transaction costs can be very important for asset pricing when an investment opportunity set is stochastic. We also show that concerns over a potential liquidity crash, no matter how unlikely, can dramatically reduce investment in stock, even when the current market is perfectly liquid and the expected excess return is high. Our analysis thus suggests that a stochastic investment opportunity set with small transaction costs and/or the existence of liquidity risk may help partially explain the Equity-Premium Puzzle and other anomalies such as the well-documented limited participation phenomenon.

APPENDIX

In this Appendix, we first provide results on the computation of the steady-state distribution, the expected holding period, and the expected lifetime transaction costs, and then we present proofs for the propositions and theorems in this paper.

A.1 Steady-state distribution, expected holding period, and expected lifetime transaction costs

A.1.1 Steady-state distribution

It can be verified that

$$dz_t = \mu_{zi}(z_t)dt - \sigma_i z_t dw_t, \quad (\text{A-1})$$

where

$$\mu_{zi}(z) = (r_i - \mu_i + \sigma_i^2)z - \left(\frac{\psi'_i(z)}{1-k} \right)^{-1/\gamma}. \quad (\text{A-2})$$

For simplicity, we focus on the case with separated no-transaction regions. The corresponding results for other cases can be derived using the same method. We have the following result for computing the steady-state distribution of z_t .

Proposition 4. *Suppose $\underline{z}_B < \bar{z}_B < \underline{z}_b < \bar{z}_b$. Let $\phi_i(z)$ be the steady-state density function.*

Then we have

$$\phi(z) = \begin{cases} \phi_B(z) & \underline{z}_B < z < \bar{z}_B \\ \phi_b(z) & \underline{z}_b < z < \bar{z}_b \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A-3})$$

where $\phi_B(z)$ and $\phi_b(z)$ solve

$$\frac{1}{2}\sigma_B^2 z^2 \phi_B''(z) - (\mu_{zB}(z) - 2\sigma_B^2 z) \phi_B'(z) - (\lambda_B - \sigma_B^2 + \mu'_{zB}(z)) \phi_B(z) = 0,$$

subject to

$$\begin{aligned}\frac{1}{2}\sigma_B^2\bar{z}_B^2\phi'_B(\underline{z}_B) - (\mu_{zB}(\underline{z}_B) - \sigma_B^2\bar{z}_B)\phi_B(\underline{z}_B) &= 0, \\ \frac{1}{2}\sigma_B^2\bar{z}_B^2\phi'_B(\bar{z}_B) - (\mu_{zB}(\bar{z}_B) - \sigma_B^2\bar{z}_B)\phi_B(\bar{z}_B) - \frac{\lambda_b\lambda_B}{\lambda_b + \lambda_B} &= 0\end{aligned}$$

and

$$\frac{1}{2}\sigma_b^2 z^2 \phi_b''(z) - (\mu_{zb}(z) - 2\sigma_b^2 z)\phi_b'(z) - (\lambda_b - \sigma_b^2 + \mu'_{zb}(z))\phi_b(z) = 0,$$

subject to

$$\begin{aligned}\frac{1}{2}\sigma_b^2 \underline{z}_b^2 \phi_b'(\underline{z}_b) - (\mu_{zb}(\underline{z}_b) - \sigma_b^2 \underline{z}_b)\phi_b(\underline{z}_b) + \frac{\lambda_b\lambda_B}{\lambda_b + \lambda_B} &= 0, \\ \frac{1}{2}\sigma_b^2 \bar{z}_b^2 \phi_b'(\bar{z}_b) - (\mu_{zb}(\bar{z}_b) - \sigma_b^2 \bar{z}_b)\phi_b(\bar{z}_b) &= 0.\end{aligned}$$

Proof: See Subsection A.2 in this Appendix.

Note that Proposition 4. implies that

$$\int_{\underline{z}_B}^{\bar{z}_B} \phi_B(z)dz = \frac{\lambda_b}{\lambda_b + \lambda_B} \quad \text{and} \quad \int_{\underline{z}_b}^{\bar{z}_b} \phi_b(z)dz = \frac{\lambda_B}{\lambda_b + \lambda_B}.$$

A.1.2 Expected holding period

Next, to investigate the frequency of trade in our model, we now show how to compute the expected time to the next sale from a given initial position. We focus on the case where the no-transactions regions are separated.

In this case, we can define the next sale time to be

$$\tau_s = \inf\{t \geq 0 : z_t = \underline{z}_B \text{ or } z_t = \underline{z}_b\}$$

and the expected time to the next sale starting from z in Regime i to be

$$T_i(z) = E[\tau_s | z_0 = z].$$

The following proposition provides results on computing the expected time to the next sale.

Proposition 5. *Suppose $\underline{z}_B < \bar{z}_B < \underline{z}_b < \bar{z}_b$. Then*

(i) $T_B(z)$ solves

$$\frac{1}{2}\sigma_B^2 z^2 T_B''(z) + \mu_{zB}(z)T_B'(z) - \lambda_B T_B(z) + 1 = 0,$$

with boundary conditions

$$T_B(\underline{z}_B) = 0, \quad T_B'(\bar{z}_B) = 0;$$

(ii) $T_b(z)$ solves

$$\frac{1}{2}\sigma_b^2 z^2 T_b''(z) + \mu_{zb}(z)T_b'(z) + \lambda_b(T_B(\bar{z}_B) - T_b(z)) + 1 = 0,$$

with boundary conditions

$$T_b(\underline{z}_b) = 0, \quad T_b'(\bar{z}_b) = 0,$$

where μ_{zi} is as defined in (A-2).

Proof: See Subsection A.2 in this Appendix.

A.1.3 Expected transaction costs

Another measure of the effect of transaction costs is the expected discounted transaction costs an investor expects to pay over the entire investment horizon. The following propo-

sition shows one way of computing these costs for the case with separated no-transaction regions.

Proposition 6. *Suppose $\nu > \max_i[-\lambda_i + r_i + (\mu_i - r_i)/(\underline{z}_B + 1)]$ and $\underline{z}_B < \bar{z}_B < \underline{z}_b < \bar{z}_b$. Let $C_i(x_t, y_t)$ ($i \in \{B, b\}$) be the expected discounted transaction costs in regime i starting from (x_t, y_t) . Then given the optimal polity (I_t^*, D_t^*) ,*

$$C_i(x, y) \equiv E \left[\int_0^{\tau_i} e^{-\nu t} (\theta_i dI_t^* + \alpha_i dD_t^*) + e^{-\nu \tau_i} C_j(x_{\tau_i}, y_{\tau_i}) \right] = y g_i(x/y),$$

where for $\underline{z}_i \leq z \leq \bar{z}_i$, $g_i(\cdot)$ solves

$$\frac{1}{2} \sigma_i^2 z^2 g_i''(z) - \left((\mu_i - r_i)z + \left(\frac{\psi'(z)}{1-k} \right)^{-1/\gamma} \right) g_i'(z) - (\nu + \lambda_i - \mu_i) g_i(z) + \lambda_i g_j(z) = 0, \quad (\text{A-4})$$

subject to

$$g_i(\bar{z}_i) - (\bar{z}_i + 1 + \theta_i) g_i'(\bar{z}_i) + \theta_i = 0 \quad (\text{A-5})$$

and

$$g_i(\underline{z}_i) - (\underline{z}_i + 1 - \alpha_i) g_i'(\underline{z}_i) - \alpha_i = 0, \quad (\text{A-6})$$

$$g_B(z) = \frac{1}{\bar{z}_B + 1 + \theta_B} ((z + 1 + \theta_B) g_B(\bar{z}_B) + (z - \bar{z}_B) \theta_B) \quad (\text{A-7})$$

for $z > \bar{z}_B$, and

$$g_b(z) = \frac{1}{\underline{z}_b + 1 - \alpha_b} ((z + 1 - \alpha_b) g_b(\underline{z}_b) + (\underline{z}_b - z) \alpha_b) \quad (\text{A-8})$$

for $z < \underline{z}_b$.

Proof: See Subsection A.2 in this Appendix.

A.2 Proofs

PROOF OF LEMMA 1.. We prove for the case with $\eta_i < \eta_j$. The other case can be proved using a similar argument. (10) implies that

$$M_j = f(M_i) = \frac{\eta_i + \lambda_i}{\lambda_i} M_i - \frac{\gamma}{\lambda_i} (1 - k)^{1/\gamma} M_i^{1-1/\gamma} - \frac{k\delta}{\lambda_i}.$$

For $i \in \{B, b\}$, let $\bar{M}_i > 0$ be the unique solution of

$$-\eta_i M_i + \gamma(1 - k)^{1/\gamma} (M_i)^{1-1/\gamma} + k\delta = 0$$

and let $\underline{M}_i > 0$ be the unique solution of

$$-(\eta_i + \lambda_i) M_i + \gamma(1 - k)^{1/\gamma} (M_i)^{1-1/\gamma} + k\delta = 0.$$

Define

$$g(M_i) = -(\eta_j + \lambda_j) f(M_i) + \gamma(1 - k)^{1/\gamma} f(M_i)^{1-1/\gamma} + \lambda_j M_i + k\delta.$$

After simplification, we have

$$g(\bar{M}_i) = (\eta_i - \eta_j) \bar{M}_i < 0.$$

In addition, it can be easily verified that $g(\underline{M}_i) = \lambda_j \underline{M}_i + k\delta \geq 0$. By continuity, there exists an M_i such that $g(M_i) = 0$. This implies that $\bar{M}_i \geq M_i \geq \underline{M}_i$. Finally, if $\gamma < 1$, it is obvious that $g'(x) > 0$, $\forall x > 0$ and therefore the solution is unique. If $\gamma > 1$, direct computation reveals that $g''(x) < 0$, $g'(0) > 0$, and $g(\underline{M}_i) > 0$, which also implies that the solution is unique. $\square$

PROOF OF THEOREM 2.. We first state some properties of the candidate value function that satisfies the conditions in Theorem 2..

Lemma 2. *Suppose $v_i(x, y)$ and $v_j(x, y)$ are as in Theorem 2.. Then we have*

1. $v_i(x, y) \geq v_i(x + (1 - \alpha_i)y, 0)$.

2. *There exist k_i and K_j such that*

$$k_i(x + y)^{1-\gamma} \leq v_i(x, y) \leq K_i(x + y)^{1-\gamma}. \quad (\text{A-9})$$

3. $\frac{yv_{iy}}{v_i}$ is bounded for $\frac{x}{y} > \alpha_i - 1$.

4. $\frac{v_{ix}^{\frac{\gamma-1}{\gamma}}}{v_i}$ is bounded for $\frac{x}{y} > \alpha_i - 1$.

PROOF OF LEMMA 2.. Part 1 follows from the well known inequality for concave functions U

$$\nabla U(z_1) \cdot (z_1 - z_2) \leq U(z_1) - U(z_2). \quad (\text{A-10})$$

$$[v_{ix}(x, y) \quad v_{iy}(x, y)][x - (x + (1 - \alpha_i)y) \quad y]^\top \leq v_i(x, y) - v_i(x + (1 - \alpha_i)y, 0). \quad (\text{A-11})$$

Using condition (21), we have

$$0 \leq y(v_{iy}(x, y) - (1 - \alpha_i)v_{ix}(x, y)) \leq v_i(x, y) - v_i(x + (1 - \alpha_i)y, 0), \quad (\text{A-12})$$

which proves 1.

Part 2 follows from the fact that $v_i(x, y) = (x + y)^{1-\gamma}v_i(\frac{x}{x+y}, \frac{y}{x+y})$.

Part 3 follows from $v_i(x, y) = y^{1-\gamma}\phi(\frac{x}{y})$, $v_{iy} = (1 - \gamma)y^{-\gamma}\phi(\frac{x}{y}) - y^{-\gamma}\frac{x}{y}\phi'(\frac{x}{y})$ so $\frac{yv_{iy}}{v_i} = (1 - \gamma) - \frac{\frac{x}{y}\phi'(\frac{x}{y})}{\phi(\frac{x}{y})}$, which is bounded.

Part 4 follows from $v_{ix} = y^{-\gamma} \phi'(\frac{x}{y})$, $v_{ix}^{1-\frac{1}{\gamma}} = y^{1-\gamma} (\phi'(\frac{x}{y}))^{1-\frac{1}{\gamma}}$ so $\frac{v_{ix}^{\frac{\gamma-1}{\gamma}}}{v_i} = \frac{(\phi'(\frac{x}{y}))^{1-\frac{1}{\gamma}}}{\phi(\frac{x}{y})}$, which is bounded. $\square$

Now we are ready to prove Theorem 2..

The proof relies on results first proved in Davis and Norman (1990). We repeat many of their arguments here, adapting for our particular setting. Applying Itô's lemma to $\log[e^{-(\rho+\delta+\lambda_i)t}(1-\gamma)v_i(x(t), y(t))]$ leads to:

$$\begin{aligned}
& e^{-(\rho+\delta+\lambda_i)t} v_i(x_t, y_t) \\
&= v_i(x, y) \exp\left(\int_0^t \frac{1}{v_i} \left(Gv - (1-k) \frac{c_s^{1-\gamma}}{1-\gamma} - k\delta \frac{(x_s + (1-\alpha_i)y_s)^{1-\gamma}}{1-\gamma} - \lambda_i v_j\right) ds\right. \\
&+ \int_0^t \frac{1}{v_i} [(v_{iy} - (1+\theta_i)v_{ix})dI_s + ((1-\alpha_i)v_{ix} - v_{iy})dD_s] \\
&+ \left. \int_0^t \frac{1}{v_i} v_{iy} y \sigma dw_s - \frac{1}{2} \int_0^t \frac{v_{iy}^2}{v_i^2} \sigma^2 y^2 ds\right), \tag{A-13}
\end{aligned}$$

where $Gv \equiv \frac{1}{2}\sigma^2 y^2 v_{yy} + r_i x v_{ix} - c v_{ix} + \mu_i y v_{iy} - (\rho + \delta)v_i + \lambda_i(v_j - v_i) + (1-k)\frac{c^{1-\gamma}}{1-\gamma} + k\delta \frac{(x+(1-\alpha_i)y)^{1-\gamma}}{1-\gamma}$.

In particular, setting $c = c^* \equiv \left(\frac{v_{ix}}{1-k}\right)^{-\frac{1}{\gamma}}$ and following the candidate transaction policy, we have $Gv \equiv 0$ in the no-transaction cost region, the terms involving dI and dD are 0, and $\frac{1}{v_i}((1-k)\frac{c^{1-\gamma}}{1-\gamma} + k\delta \frac{(x+(1-\alpha_i)y)^{1-\gamma}}{1-\gamma} + \lambda_i v_j)$ is a positive bounded function bounded away from 0. Moreover, $\frac{v_{iy}y}{v_i}$ is a bounded function for the candidate transaction policy. These properties are proved in Lemma 2.. Notice that using Itô's Lemma we also have for a sequence of stopping times $\tau_n \rightarrow \infty$

$$\begin{aligned}
v_i(x, y) &= E\left[\int_0^{\tau_n \wedge t} e^{-(\rho+\delta+\lambda_i)s} \left((1-k) \frac{c_s^{*1-\gamma}}{1-\gamma} + k\delta \frac{(x_s^* + (1-\alpha_i)y_s^*)^{1-\gamma}}{1-\gamma} + \lambda_i v_j(x_s^*, y_s^*)\right) ds\right. \\
&\quad \left.+ e^{-(\rho+\delta+\lambda_i)\tau_n \wedge t} v_i(x_{\tau_n \wedge t}^*, y_{\tau_n \wedge t}^*)\right]. \tag{A-14}
\end{aligned}$$

From (A-13) we see if $\gamma > 1$

$$0 \geq v_i(x_{\tau_n \wedge t}^*, y_{\tau_n \wedge t}^*) \geq v_i(x, y) \exp\left(\int_0^{\tau_n \wedge t} \frac{1}{v_i} v_{iy} y_s^* \sigma dw_s - \frac{1}{2} \int_0^{\tau_n \wedge t} \frac{v_{iy}^2}{v_i^2} \sigma^2 y_s^{*2} ds\right) e^{(\rho + \delta + \lambda_i)t} \quad (\text{A-15})$$

while if $0 < \gamma < 1$

$$0 \leq v_i(x_{\tau_n \wedge t}^*, y_{\tau_n \wedge t}^*) \leq v_i(x, y) \exp\left(\int_0^{\tau_n \wedge t} \frac{1}{v_i} v_{iy} y_s^* \sigma dw_s - \frac{1}{2} \int_0^{\tau_n \wedge t} \frac{v_{iy}^2}{v_i^2} \sigma^2 y_s^{*2} ds\right) e^{(\delta + \lambda_i)t} \quad (\text{A-16})$$

We remind the reader that the exponential local martingales in Equations (A-15) and (A-16) are in fact Class D martingales since $\frac{1}{v_i} v_{iy} y$ is bounded. Letting $n \rightarrow \infty$ in Equation (A-14), observe that random variables $v_i(x_{\tau_n \wedge t}^*, y_{\tau_n \wedge t}^*)$ are bounded by uniformly integrable random variables and using the dominated convergence theorem (Royden (1988) Proposition 18, p. 270)

$$\begin{aligned} v_i(x, y) &= E\left[\int_0^t e^{-(\rho + \delta + \lambda_i)s} \left((1 - k) \frac{c_s^{*1-\gamma}}{1 - \gamma} + k \delta \frac{(x_s^* + (1 - \alpha_i) y_s^*)^{1-\gamma}}{1 - \gamma} + \lambda_i v_j(x_s^*, y_s^*)\right) ds \right. \\ &\quad \left. + e^{-(\rho + \delta + \lambda_i)t} v_i(x_t^*, y_t^*)\right]. \end{aligned} \quad (\text{A-17})$$

We also have, using (A-13)

$$\begin{aligned} &e^{-(\rho + \delta + \lambda_i)t} v_i(x_t, y_t) \\ &= v_i(x, y) \exp\left(\int_0^t \frac{1}{v_i} \left(-(1 - k) \frac{c_s^{*1-\gamma}}{1 - \gamma} - k \delta \frac{(x_s^* + (1 - \alpha_i) y_s^*)^{1-\gamma}}{1 - \gamma} - \lambda_i v_j\right) ds\right) \\ &\times \exp\left(\int_0^t \frac{1}{v_i} v_{iy} y \sigma dw_s - \frac{1}{2} \int_0^t \frac{v_{iy}^2}{v_i^2} \sigma^2 y^2 ds\right). \end{aligned} \quad (\text{A-18})$$

Since $\frac{1}{v_i} \left(-(1 - k) \frac{c_s^{*1-\gamma}}{1 - \gamma} - k \delta \frac{(x_s^* + (1 - \alpha_i) y_s^*)^{1-\gamma}}{1 - \gamma} - \lambda_i v_j\right)$ is a negative function bounded away from 0, and the exponential local martingale is a Class D martingale, we have

$\lim_{t \rightarrow \infty} e^{(-\delta + \lambda_i)t} E[v_i(x^*(t), y^*(t))] = 0$. As a result,

$$v_i(x, y) = E\left[\int_0^\infty e^{-(\rho + \delta + \lambda_i)s} \left((1 - k) \frac{c_s^{1-\gamma}}{1 - \gamma} + k\delta \frac{(x_s^* + (1 - \alpha_i)y_s^*)^{1-\gamma}}{1 - \gamma} + \lambda_i v_j(x_s^*, y_s^*)\right) ds\right]. \quad (\text{A-19})$$

Next we show that given v_j , v_i is the value function and vice versa. We start by considering the case $\gamma > 1$. Consider trading strategies that start with $(x + \epsilon, y)$, and follow an admissible consumption and trading strategy for initial endowments (x, y) , say $(c, \hat{x}, \hat{y}) \in \Theta(x, y)$ and maintain ϵe^{rt} in the risk-free account. For these strategies, a simple application of Itô's lemma for a set of stopping times $\tau_n \rightarrow \infty$ leads to

$$\begin{aligned} & v_i(x + \epsilon, y) \\ \geq & E\left[\int_0^{\tau_n \wedge t} e^{-(\rho + \delta + \lambda_i)s} \left((1 - k) \frac{c_s^{1-\gamma}}{1 - \gamma} + k\delta \frac{(x_s + (1 - \alpha_i)y_s)^{1-\gamma}}{1 - \gamma} + \lambda_i v_j(\hat{x}_s + \epsilon e^{rs}, \hat{y}_s)\right) ds \right. \\ & \left. + e^{-(\rho + \delta + \lambda_i)\tau_n \wedge t} v_i(\hat{x}_{\tau_n \wedge t} + \epsilon e^{r\tau_n \wedge t}, \hat{y}_{\tau_n \wedge t})\right]. \end{aligned} \quad (\text{A-20})$$

From monotonicity and Part 1 of Lemma 2.,

$$\begin{aligned} 0 & \geq e^{-(\rho + \delta + \lambda_i)\tau_n \wedge t} v_i(\hat{x}_{\tau_n \wedge t} + \epsilon e^{r\tau_n \wedge t}, \hat{y}_{\tau_n \wedge t}) \geq e^{-(\rho + \delta + \lambda_i)\tau_n \wedge t} v_i(\hat{x}_{\tau_n \wedge t} - (1 - \alpha_i)\hat{y}_{\tau_n \wedge t} + \epsilon e^{r\tau_n \wedge t}, 0) \\ & \geq e^{-(\rho + \delta + \lambda_i)\tau_n \wedge t} v_i(\epsilon, 0) \end{aligned}$$

and so by the dominated convergence theorem, we can let $n \rightarrow \infty$ to get

$$\begin{aligned} v_i(x + \epsilon, y) & \geq E\left[\int_0^t e^{-(\rho + \delta + \lambda_i)s} \left((1 - k) \frac{c_s^{1-\gamma}}{1 - \gamma} + k\delta \frac{(x_s + (1 - \alpha_i)y_s)^{1-\gamma}}{1 - \gamma} + \lambda_i v_j(\hat{x}_s + \epsilon e^{rs}, \hat{y}_s)\right) ds \right. \\ & \left. + e^{-(\rho + \delta + \lambda_i)t} v_i(\hat{x}_t + \epsilon e^{rt}, \hat{y}_t)\right]. \end{aligned} \quad (\text{A-21})$$

Letting $t \rightarrow \infty$ we have $0 \geq e^{-(\rho + \delta + \lambda_i)t} v_i(\hat{x}_t + \epsilon e^{rt}, \hat{y}_t) \geq e^{-(\rho + \delta + \lambda_i)t} v_i(\epsilon, 0) \rightarrow 0$. Using

the monotone convergence theorem, we then have

$$v_i(x + \epsilon, y) \geq E\left[\int_0^\infty e^{-(\rho+\delta+\lambda_i)s} \left(\frac{c_s^{1-\gamma}}{1-\gamma} + \lambda_i v_j(\hat{x}_s + \epsilon e^{rs}, \hat{y}_s)\right) ds\right]. \quad (\text{A-22})$$

Next, letting $\epsilon \downarrow 0$ using the continuity of v_i and the monotone convergence theorem, we have

$$v_i(x, y) \geq E\left[\int_0^\infty e^{-(\rho+\delta+\lambda_i)s} \left((1-k) \frac{c_s^{1-\gamma}}{1-\gamma} - k\delta \frac{(x_s + (1-\alpha_i)y_s)^{1-\gamma}}{1-\gamma} + \lambda_i v_j(x_s, y_s)\right) ds\right] \quad (\text{A-23})$$

for all feasible consumption trading strategies in $\Theta(x, y)$. This implies v is the value function given v_j .

In the case $0 < \gamma < 1$, we have

$$\begin{aligned} v_i(x, y) \geq & E\left[\int_0^{\tau_n \wedge t} e^{-(\rho+\delta+\lambda_i)s} \left((1-k) \frac{c_s^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_s + (1-\alpha_i)y_s)^{1-\gamma}}{1-\gamma} + \lambda_i v_j(x_s, y_s)\right) ds \right. \\ & \left. + e^{-(\rho+\delta+\lambda_i)\tau_n \wedge t} v_i(x_{\tau_n \wedge t}, y_{\tau_n \wedge t})\right] \end{aligned} \quad (\text{A-24})$$

and $v_i(x_t, y_t) \geq 0$. This leads immediately to the conclusion

$$v_i(x, y) \geq E\left[\int_0^\infty e^{-(\rho+\delta+\lambda_i)s} \left((1-k) \frac{c_s^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_s + (1-\alpha_i)y_s)^{1-\gamma}}{1-\gamma} + \lambda_i v_j(x_s, y_s)\right) ds\right] \quad (\text{A-25})$$

for all feasible consumption trading strategies from the initial position (x, y) . This implies v_i is the value function given v_j . Similar argument shows that v_j is the value function given v_i .

Finally, we show that v_i and v_j are the values in Regimes i and j respectively. First,

we define

$$v_i(x, y) = \sup_{(c, D, I) \in \Theta(x, y)} E \left[\int_0^\infty e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt \right]. \quad (\text{A-26})$$

Let $v_n(x, y) = v_j(x, y)$ if n is odd and $v_n(x, y) = v_i(x, y)$ if n is even. Let τ_n be the time of the n th regime change. The above proof implies

$$v_i(x, y) = E \left[\int_0^{\tau_1} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt + e^{-(\rho+\delta)\tau_1} v_1(x_{\tau_1}^*, y_{\tau_1}^*) \right] \quad (\text{A-27})$$

and for all feasible consumption-investment strategies

$$v_i(x, y) \geq E \left[\int_0^{\tau_1} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt + e^{-(\rho+\delta)\tau_1} v_1(x_{\tau_1}, y_{\tau_1}) \right] \quad (\text{A-28})$$

Given time τ_1 information, we also know

$$\begin{aligned} v_1(x_{\tau_1}, y_{\tau_1}) &= E_{\tau_1} \left[\int_{\tau_1}^{\tau_2} e^{-(\rho+\delta)(t-\tau_1)} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t^* + (1-\alpha_i)y_t^*)^{1-\gamma}}{1-\gamma} \right) dt \right. \\ &\quad \left. + e^{-(\rho+\delta)(\tau_2-\tau_1)} v_2(x_{\tau_2}^*, y_{\tau_2}^*) \right] \end{aligned} \quad (\text{A-29})$$

and for all feasible consumption-investment strategies

$$\begin{aligned} v_1(x_{\tau_1}, y_{\tau_1}) &\geq E \left[\int_{\tau_1}^{\tau_2} e^{-(\rho+\delta)(t-\tau_1)} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt \right. \\ &\quad \left. + e^{-(\rho+\delta)(\tau_2-\tau_1)} v_2(x_{\tau_2}, y_{\tau_2}) \right]. \end{aligned} \quad (\text{A-30})$$

Inserting these expressions into Equations (A-27) and (A-28) yields

$$v_i(x, y) = E \left[\int_0^{\tau_2} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt + e^{-(\rho+\delta)\tau_2} v_2(x_{\tau_2}^*, y_{\tau_2}^*) \right] \quad (\text{A-31})$$

and

$$v_i(x, y) \geq E\left[\int_0^{\tau_2} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt + e^{-(\rho+\delta)\tau_2} v_2(x_{\tau_2}, y_{\tau_2}) \right]. \quad (\text{A-32})$$

Continuing in this manner, we have

$$v_i(x, y) = E\left[\int_0^{\tau_n} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t^* + (1-\alpha_i)y_t^*)^{1-\gamma}}{1-\gamma} \right) dt + e^{-(\rho+\delta)\tau_n} v_n(x_{\tau_n}^*, y_{\tau_n}^*) \right] \quad (\text{A-33})$$

and

$$v_i(x, y) \geq E\left[\int_0^{\tau_n} e^{-(\rho+\delta)t} \left((1-k) \frac{c_s^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_s + (1-\alpha_i)y_s)^{1-\gamma}}{1-\gamma} \right) ds + e^{-(\rho+\delta)\tau_n} v_n(x_{\tau_n}, y_{\tau_n}) \right]. \quad (\text{A-34})$$

We now consider the case $\gamma > 1$. As in the previous theorem, consider strategies that start with an initial position $(x + \epsilon, y)$, but follow a feasible consumption and trading strategy for an initial position (x, y) and always maintain ϵe^{rt} in the riskless account. Very similar arguments to those in Theorem 2.1 of Davis and Norman (1990) lead to the conclusion

$$v_i(x + \epsilon, y) \geq E\left[\int_0^\infty e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt \right], \quad (\text{A-35})$$

and letting $\epsilon \rightarrow 0$, we have

$$v_i(x, y) \geq E\left[\int_0^\infty e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt \right] \quad (\text{A-36})$$

for all feasible consumption and trading strategies.

Since $v_n < 0$ from Equation (A-33), it follows

$$v_i(x, y) \leq E\left[\int_0^{\tau_n} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt\right] \quad (\text{A-37})$$

and since $\tau_n \rightarrow \infty$ almost surely as $n \rightarrow \infty$, we have from the monotone convergence theorem,

$$v_i(x, y) \leq E\left[\int_0^{\infty} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{*1-\gamma}}{1-\gamma} + k\delta \frac{(x_t^* + (1-\alpha_i)y_t^*)^{1-\gamma}}{1-\gamma} \right) dt\right] \quad (\text{A-38})$$

Thus from Equation (A-36) and Equation (A-38), we have

$$v_i(x, y) = E\left[\int_0^{\infty} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{*1-\gamma}}{1-\gamma} + k\delta \frac{(x_t^* + (1-\alpha_i)y_t^*)^{1-\gamma}}{1-\gamma} \right) dt\right] \quad (\text{A-39})$$

and

$$v_i(x, y) \geq E\left[\int_0^{\infty} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt\right] \quad (\text{A-40})$$

for all feasible trading and consumption strategies. We have proved the result for $\gamma > 1$.

When $0 < \gamma < 1$, $v_n > 0$. From Equation (A-34), we have

$$v_i(x, y) \geq E\left[\int_0^{\infty} e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{1-\gamma}}{1-\gamma} + k\delta \frac{(x_t + (1-\alpha_i)y_t)^{1-\gamma}}{1-\gamma} \right) dt\right] \quad (\text{A-41})$$

for all feasible consumption and trading strategies. To conclude the proof, we need to show that

$$\lim_{n \rightarrow \infty} E[e^{-(\rho+\delta)\tau_n} v_n(x_{\tau_n}^*, y_{\tau_n}^*)] = 0 \quad (\text{A-42})$$

since this and Equation (A-33) would imply

$$v_i(x, y) = E\left[\int_0^\infty e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{*1-\gamma}}{1-\gamma} + k\delta \frac{(x_t^* + (1-\alpha_i)y_t^*)^{1-\gamma}}{1-\gamma} \right) dt\right]. \quad (\text{A-43})$$

Equation (A-42) follows from the observation that

$$\lim_{\tau_n \rightarrow \infty} E\left[\int_{\tau_n}^\infty e^{-(\rho+\delta)t} \left((1-k) \frac{c_t^{*1-\gamma}}{1-\gamma} + k\delta \frac{(x_t^* + (1-\alpha_i)y_t^*)^{1-\gamma}}{1-\gamma} \right) dt\right] \rightarrow 0,$$

so

$$\lim_{n \rightarrow \infty} E\left[e^{-(\rho+\delta)\tau_n} \left((1-k) \frac{c_{\tau_n}^{*1-\gamma}}{1-\gamma} + k\delta \frac{(x_{\tau_n}^* + (1-\alpha_i)y_{\tau_n}^*)^{1-\gamma}}{1-\gamma} \right)\right] = 0$$

and

$$0 \leq v_n(x, y) \leq K \left((1-k) \frac{c_{\tau_n}^{*1-\gamma}}{1-\gamma} + k\delta \frac{(x_{\tau_n}^* + (1-\alpha_i)y_{\tau_n}^*)^{1-\gamma}}{1-\gamma} \right) \quad (\text{A-44})$$

for a suitable constant K , which can be derived from the homotheticity properties of v_i and v_j (implied by Parts 2 and 4 in Lemma 2.). $\square$

Lemma 3. *Suppose $\phi(x, y)$ satisfies*

$$(1-\alpha)\phi_x \leq \phi_y \leq (1+\theta)\phi_x, \quad (\text{A-45})$$

(x_t, y_t) correspond to the stock and bond accounts using the optimal c, I, D in regime i ,

$$\lim_{T \rightarrow \infty} E[e^{-(\rho+\lambda_i)T} \phi(x_T, y_T)] = 0, \quad (\text{A-46})$$

and $E[\int_0^T \phi_y y_t^2 \sigma_i^2 dw_t] = 0$. Let $\mathcal{S}_1$ be the set of $(x, y) \in \mathcal{S}$ such that

$$0 > \frac{1}{2} \sigma_i^2 y^2 \phi_{yy} + r x \phi_x + \mu_i y \phi_y - \rho \phi + \frac{\gamma}{1-\gamma} (\phi_x)^{1-\frac{1}{\gamma}} + \lambda_i (v_j - \phi), \quad (\text{A-47})$$

and let $\mathcal{S}_2 = \{(x, y) \in \mathcal{S} | \phi(x, y) \geq v_i(x, y)\}$. Then if $\mathcal{S}_1 \cup \mathcal{S}_2 = \mathcal{S}$, $\phi(x, y) \geq v_i(x, y)$ for all $(x, y) \in \mathcal{S}$.

Proof. We just need to consider $(x, y) \in \mathcal{S}_1$. Let (x_t, y_t) correspond to the optimal bond and stock account using the optimal c, D, I in regime i , and let $\tau = \inf\{t | \frac{1}{2}\sigma_i^2 y_t^2 \phi_{yy} + r x_t \phi_x + \mu_i y_t \phi_y - \rho \phi + \frac{\gamma}{1-\gamma}(\phi_x)^{1-\frac{1}{\gamma}} + \lambda_i(v_j - \phi) \geq 0\}$. In other words, $\tau = \inf\{t | (x_t, y_t) \in \mathcal{S}_2 \setminus \mathcal{S}_1\}$.

By Ito's lemma we have

$$\begin{aligned}
& e^{-(\rho+\lambda_i)T \wedge \tau} \phi(x_{T \wedge \tau}, y_{T \wedge \tau}) + \int_0^{T \wedge \tau} e^{-(\rho+\lambda_i)t} \left(\frac{c_t^{1-\gamma}}{1-\gamma} + \lambda_i v_j(x_t, y_t) \right) dt \\
= & \phi(x_0, y_0) + \int_0^{T \wedge \tau} e^{-(\rho+\lambda_i)t} \left(\frac{1}{2} \sigma_i^2 y_t^2 \phi_{yy} + r x_t \phi_x + \mu_i y_t \phi_y - c_t \phi_x + \frac{c_t^{1-\gamma}}{1-\gamma} - \rho \phi \right. \\
& \left. + \lambda_i(v_j - \phi) \right) dt + \int_0^{T \wedge \tau} e^{-(\rho+\lambda_i)t} (\phi_y - (1+\theta)\phi_x) dI_t \\
& + \int_0^{T \wedge \tau} e^{-(\rho+\lambda_i)t} ((1-\alpha)\phi_x - \phi_y) dD_t + \int_0^{T \wedge \tau} e^{-(\rho+\lambda_i)t} \sigma_i y_t \phi_y dw_t. \tag{A-48}
\end{aligned}$$

We have $-c_t \phi_x + \frac{c_t^{1-\gamma}}{1-\gamma} \leq \frac{\gamma}{1-\gamma}(\phi_x)^{1-\frac{1}{\gamma}}$. Taking expectations, we have

$$\begin{aligned}
\phi(x, y) \geq & E \left[\int_0^{T \wedge \tau} e^{-(\rho+\lambda_i)t} \left(\frac{c_t^{1-\gamma}}{1-\gamma} + \lambda_i v_j(x_t, y_t) \right) dt + e^{-(\rho+\lambda_i)T} \phi(x_T, y_T) \mathbf{1}_{\{T \leq \tau\}} \right. \\
& \left. + e^{-(\rho+\lambda_i)\tau} v_i(x_\tau, y_\tau) \mathbf{1}_{\{\tau < T\}} \right]
\end{aligned}$$

Letting $T \rightarrow \infty$ gives

$$\phi(x, y) \geq E \left[\int_0^{\infty \wedge \tau} e^{-(\rho+\lambda_i)t} \left(\frac{c_t^{1-\gamma}}{1-\gamma} + \lambda_i v_j(x_t, y_t) \right) dt + e^{-(\rho+\lambda_i)\tau} v_i(x_\tau, y_\tau) \mathbf{1}_{\{\tau < \infty\}} \right] = v_i(x, y). \tag{A-49}$$

□

PROOF OF PROPOSITION 1. Proof of statement 1. The left hand inequality simply expresses the fact that the investor cannot be worse off than as following the feasible

strategy of liquidating the stock and investing all wealth in the money market and optimally consuming. To establish the right hand inequality, we follow the strategy of Shreve and Soner (1994), Theorem 9.9. Consider the function $\phi(x, y) = \frac{A}{1-\gamma}(x + \zeta y)^{1-\gamma}$, where $1 - \alpha \leq \zeta \leq 1 + \theta$. We have $(1 - \alpha)\phi_x \leq \phi_y \leq (1 + \theta)\phi_x$. Moreover, evaluating the PDE (13),

$$\frac{1}{2}\sigma_i^2 y^2 \phi_{yy} + rx\phi_x + \mu_i y \phi_y - \rho\phi + \frac{\gamma}{1-\gamma}(\phi_x)^{1-\frac{1}{\gamma}} + \lambda_i(v_j - \phi) \quad (\text{A-50})$$

is equal to

$$\begin{aligned} & -A(x + \zeta y)^{1-\gamma} \times \left[\frac{\rho + \lambda_i - (1-\gamma)(r + \frac{\kappa_i^2}{2\gamma})}{1-\gamma} + \left(\sqrt{\frac{\gamma}{2}}\sigma_i \frac{\zeta y}{x + \zeta y} - \kappa_i \sqrt{\frac{1}{2\gamma}} \right)^2 \right. \\ & \left. - \frac{\gamma A^{-\frac{1}{\gamma}}}{1-\gamma} - \frac{\lambda_i}{A} \frac{v_j}{(x + \zeta y)^{1-\gamma}} \right], \end{aligned} \quad (\text{A-51})$$

where $\kappa_i = \frac{\mu_i - r}{\sigma_i}$. To apply the argument of Shreve and Soner (1994), we want to choose A to guarantee the bracketed term in (A-51) is nonnegative. A sufficient condition for this is

$$\frac{\rho + \lambda_i - (1-\gamma)(r + \frac{\kappa_i^2}{2\gamma})}{1-\gamma} - \frac{\gamma A^{-\frac{1}{\gamma}}}{1-\gamma} - \frac{\lambda_i}{A} \frac{v_j}{(x + \zeta y)^{1-\gamma}} \geq 0, \quad (\text{A-52})$$

and since $v_j < 0$, we can choose A so that

$$\frac{\rho + \lambda_i - (1-\gamma)(r + \frac{\kappa_i^2}{2\gamma})}{1-\gamma} - \frac{\gamma A^{-\frac{1}{\gamma}}}{1-\gamma} = 0 \quad (\text{A-53})$$

or $A = (\frac{\eta_i + \lambda_i}{\gamma})^{-\gamma} = \underline{M}_i$ where η_i is defined in (11) and $\underline{M}_i$ is defined in the proof of Lemma 1.. It follows from using Lemma 3. that $v_i(x, y) \leq \frac{\underline{M}_i}{1-\gamma}(x + \zeta y)^{1-\gamma}$. Also note that this implies $v_j(x, y) \leq \frac{\underline{M}_j}{1-\gamma}(x + \zeta y)^{1-\gamma}$. Having obtained this bound, we can improve on it by substituting equation (A-52) for v_j and producing a new bound. Iterating in this manner leads to the result.

Proof of statement 2. We have

$$0 \geq \frac{1}{2}\sigma_i^2 y^2 v_{iyy} + rxv_{ix} + \mu_i y v_{iy} - \rho v_i + \frac{\gamma}{1-\gamma}(v_{ix})^{1-\frac{1}{\gamma}} + \lambda_i(v_j - v_i), \quad (\text{A-54})$$

which can be written as

$$\begin{aligned} 0 \geq & \frac{1}{2}\sigma_j^2 y^2 v_{iyy} + rxv_{ix} + \mu_j y v_{iy} - \rho v_i + \frac{\gamma}{1-\gamma}(v_{ix})^{1-\frac{1}{\gamma}} \\ & + \frac{1}{2}(\sigma_i^2 - \sigma_j^2)y^2 v_{iyy} + (\mu_i - \mu_j)y v_{iy} + \lambda_i(v_j - v_i). \end{aligned} \quad (\text{A-55})$$

Thus, if the first line is greater than or equal to zero, $v_i(x, y) \geq v_j(x, y)$. If the first line is less than zero, then using Lemma 3. also leads to $v_i(x, y) \geq v_j(x, y)$. Using this result, we can write the PDE in regime i for $\frac{x}{y} = \underline{z}_i$ as

$$\begin{aligned} 0 \leq & -\frac{\gamma}{2}\sigma_i^2(1-\alpha)^2 + (\mu_i - r)(1-\alpha)(\underline{z}_i + 1 - \alpha) \\ & + \{-(\rho - (1-\gamma)r)\frac{1}{1-\gamma} + \frac{\gamma}{1-\gamma}A_i^{-\frac{1}{\gamma}}\}(\underline{z}_i + 1 - \alpha)^2, \end{aligned} \quad (\text{A-56})$$

so

$$0 \leq -\frac{\gamma}{2}\sigma_i^2(1-\alpha)^2 + (\mu_i - r)(1-\alpha)(\underline{z}_i + 1 - \alpha) \quad (\text{A-57})$$

from which the bound on $\underline{z}_i$ follows.

We can write the PDE in regime j for $\frac{x}{y} = \underline{z}_j$ as

$$\begin{aligned} 0 = & -\frac{\gamma}{2}\sigma_j^2(1-\alpha)^2 + (\mu_j - r)(1-\alpha)(\underline{z}_j + 1 - \alpha) + \lambda_j \frac{\psi_i(\underline{z}_j)}{A_j(\underline{z}_j + 1 - \alpha)^{-\gamma-1}} \\ & + \{-(\rho - (1-\gamma)r)\frac{1}{1-\gamma} + \frac{\gamma}{1-\gamma}A_j^{-\frac{1}{\gamma}} - \lambda_j \frac{1}{1-\gamma}\}(\underline{z}_j + 1 - \alpha)^2. \end{aligned} \quad (\text{A-58})$$

Using the bounds in statement 1, we can write this as

$$0 \leq -\frac{\gamma}{2}\sigma_j^2(1-\alpha)^2 + (\mu_j - r)(1-\alpha)(\underline{z}_j + 1 - \alpha) + (\lambda_j \frac{(\frac{M_j}{C} - 1)}{1-\gamma})(\underline{z}_j + 1 - \alpha)^2,$$

from which the bound on $\underline{z}_j$ follows.

Proof of statement 3. Suppose $\underline{z}_i < \underline{z}_j$. Then for $\frac{x}{y} = \underline{z}_i$, $v_i(x, y) = \frac{A_i}{1-\gamma}(x + (1-\alpha)y)^{1-\gamma}$ and $v_j(x, y) = \frac{A_j}{1-\gamma}(x + (1-\alpha)y)^{1-\gamma}$. Using these expressions in the PDE (13) leads to

$$\begin{aligned} 0 = & -\frac{\gamma}{2}\sigma_i^2(1-\alpha)^2 A_i + (\mu_i - r)(1-\alpha)(\underline{z}_i + 1 - \alpha)A_i \\ & + \{-(\rho - (1-\gamma)r)\frac{A_i}{1-\gamma} + \frac{\gamma}{1-\gamma}A_i^{1-\frac{1}{\gamma}} + \lambda_i \frac{A_j - A_i}{1-\gamma}\}(\underline{z}_i + 1 - \alpha)^2. \end{aligned} \quad (\text{A-59})$$

This can be written as

$$f(\underline{z}_i) + g(\underline{z}_i) = 0, \quad (\text{A-60})$$

where

$$f(z) = -A_i \left(\sqrt{\frac{\kappa_i^2}{2\gamma}}(z + 1 - \alpha) - \sigma_i(1-\alpha)\sqrt{\frac{\gamma}{2}} \right)^2 \leq 0, \quad (\text{A-61})$$

$$g(z) = \left(-\frac{(\rho - (1-\gamma)(\frac{\kappa_i^2}{2\gamma} + r))A_i}{1-\gamma} + \frac{\gamma}{1-\gamma}A_i^{1-\frac{1}{\gamma}} + \lambda_i \frac{A_j - A_i}{1-\gamma} \right) (z + 1 - \alpha)^2, \quad (\text{A-62})$$

where $\kappa_i = \frac{(\mu_i - r)}{\sigma_i}$. It also follows from PDE (13) for $z \leq \underline{z}_i$,

$$f(z) + g(z) \leq 0. \quad (\text{A-63})$$

If $f(\underline{z}_i) < 0$, then from (A-60), $g(\underline{z}_i) > 0$, which implies $g(z) > 0$ for all $z > \alpha - 1$. Thus for (A-63) to hold for all $\alpha - 1 < z \leq \underline{z}_i$, we must have

$$\underline{z}_i < (1-\alpha) \left(\frac{\gamma\sigma_i^2}{\mu_i - r} - 1 \right). \quad (\text{A-64})$$

If $f(\underline{z}_i) = 0$, then the above inequality holds with equality, and it follows $g(z) = 0$ for all z .

Proof of statement 4. This is similar to the proof of statement 3. $\square$

PROOF OF PROPOSITION 2.. In the liquid regime, since there is no transaction cost, $\psi_L(z)$ must be of the form

$$\psi_L(z) = A_L \frac{(z+1)^{1-\gamma}}{1-\gamma},$$

for some constant A_L . Since in the illiquid regime, the buying cost is infinite and the selling cost is 100%, the investor never trades in the illiquid regime. In addition, since $\lambda_I = 0$, direct computation of the objective function yields (31).

In the liquid regime, the HJB must hold at z_L^* and thus (35). The optimality of z_L^* implies that

$$A_L((\mu - r)(z_L^* + 1) - \gamma\sigma^2) \leq \lambda_L(z_L^* + 1)^{\gamma+2}\psi_I'(z_L^*) - \lambda_L(1 - \gamma)(z_L^* + 1)^{\gamma+1}\psi_I(z_L^*), \quad (\text{A-65})$$

with equality for $z_L^* < \infty$. If $\lambda_L \geq \mu - r$, it can be verified that the only solution to (35) and (A-65) is $z_L^* = \infty$ and $A_L = \gamma/(\rho - (1 - \gamma)r)$. If $\lambda_L < \mu - r$, then $z_L^* < \infty$ and z_L^* and A_L solve (35) and (36). $\square$

PROOF OF PROPOSITION 3.. The first claim follows because there is always a positive probability of the liquidity crash, and the amount in the riskless asset must be positive to satisfy the solvency constraint in the illiquid regime.

Suppose in a fictitious economy without liquidity crashes, the drift of the stock is $\hat{\mu}$ and there are no liquidity crashes and all other parameters are as before. Then the value

function is given by

$$\hat{v}(x, y) = \frac{(x + y)^{1-\gamma}}{1 - \gamma} \left(\frac{\gamma}{\rho - (1 - \gamma)(r + \frac{(\hat{\mu} - r)^2}{2\gamma\sigma^2})} \right)^\gamma \quad (\text{A-66})$$

Suppose $\mu - r > \gamma\sigma^2$. This corresponds to the case where it would be optimal to borrow in absence of transaction costs. However, it is not optimal (or even feasible) to borrow in the presence of liquidity crashes since leverage would violate the solvency constraint in the illiquid regime. Now consider the value function, $v^{NB}(x, y)$ in a the economy without liquidity crashes (no transaction costs) but where the investor is prohibited from borrowing (i.e. $x_t \geq 0$). It can be shown that the value function in this case is given by

$$v^{NB}(x, y) = \frac{(x + y)^{1-\gamma}}{1 - \gamma} \left(\frac{\gamma}{\rho - (1 - \gamma)(\mu - \frac{\gamma\sigma^2}{2})} \right)^\gamma. \quad (\text{A-67})$$

To establish a lower bound on the liquidity premium, we can solve for a $\hat{\mu}$, which equates the utility in the fictitious economy to that in the case where borrowing is prohibited.

$$\frac{(x + y)^{1-\gamma}}{1 - \gamma} \left(\frac{\gamma}{\rho - (1 - \gamma)(r + \frac{(\hat{\mu} - r)^2}{2\gamma\sigma^2})} \right)^\gamma = \frac{(x + y)^{1-\gamma}}{1 - \gamma} \left(\frac{\gamma}{\rho - (1 - \gamma)(\mu - \frac{\gamma\sigma^2}{2})} \right)^\gamma \quad (\text{A-68})$$

or

$$r + \frac{(\hat{\mu} - r)^2}{2\gamma\sigma^2} = \mu - \frac{\gamma\sigma^2}{2}. \quad (\text{A-69})$$

Solving this, we obtain

$$\hat{\mu} = r + \sigma \sqrt{2\gamma(\mu - r - \frac{\gamma\sigma^2}{2})}. \quad (\text{A-70})$$

Since the optimal policy in the economy with liquidity crashes is feasible in an economy where borrowing is prohibited, we have $v_L(x, y) \leq v^{NB}(x, y)$. Thus the liquidity premium must be larger than $\mu - \hat{\mu} = \mu - r - \sigma \sqrt{2\gamma(\mu - r - \frac{\gamma\sigma^2}{2})}$. $\square$

PROOF OF PROPOSITION 4.. Given the optimal transaction policy, any steady-state density function ϕ_i of z_t in regime i must have the form of (A-3). In addition, for any C^2 functions $f(z, B)$ and $f(z, b)$ such that $f'(\underline{z}_B, B) = f'(\bar{z}_B, B) = f'(\underline{z}_b, b) = f'(\bar{z}_b, b) = 0$, we must have

$$\begin{aligned} & \int_{\underline{z}_B}^{\bar{z}_B} \left(\frac{1}{2} \sigma_B^2 z^2 f''(z, B) + \mu_{zB}(z) f'(z, B) + \lambda_B (f(\underline{z}_b, b) - f(z, B)) \right) \phi_B(z) dz \\ & + \int_{\underline{z}_b}^{\bar{z}_b} \left(\frac{1}{2} \sigma_b^2 z^2 f''(z, b) + \mu_{zb}(z) f'(z, b) + \lambda_b (f(\bar{z}_B, B) - f(z, b)) \right) \phi_b(z) dz = 0. \end{aligned}$$

By the property of the continuous time Markov chain, we must have

$$\int_{\underline{z}_B}^{\bar{z}_B} \phi_B(z) dz = \frac{\lambda_b}{\lambda_b + \lambda_B} \quad \text{and} \quad \int_{\underline{z}_b}^{\bar{z}_b} \phi_b(z) dz = \frac{\lambda_B}{\lambda_b + \lambda_B}.$$

Then by integration by parts, we have

$$\begin{aligned} & \int_{\underline{z}_B}^{\bar{z}_B} \left[\frac{1}{2} \sigma_B^2 (z^2 \phi_B(z))'' - (\mu_{zB}(z) \phi_B(z))' - \lambda_B \phi_B(z) \right] f(z, B) dz \\ & + \int_{\underline{z}_b}^{\bar{z}_b} \left[\frac{1}{2} \sigma_b^2 (z^2 \phi_b(z))'' - (\mu_{zb}(z) \phi_b(z))' - \lambda_b \phi_b(z) \right] f(z, b) dz \\ & + f(\underline{z}_B, B) \left[\frac{1}{2} \sigma_B^2 (\underline{z}_B^2 \phi_B'(\underline{z}_B) + 2 \underline{z}_B \phi_B(\underline{z}_B)) - \mu_{zB}(\underline{z}_B) \phi_B(\underline{z}_B) \right] \\ & - f(\bar{z}_B, B) \left[\frac{1}{2} \sigma_B^2 (\bar{z}_B^2 \phi_B'(\bar{z}_B) + 2 \bar{z}_B \phi_B(\bar{z}_B)) - \mu_{zB}(\bar{z}_B) \phi_B(\bar{z}_B) - \frac{\lambda_b \lambda_B}{\lambda_b + \lambda_B} \right] \\ & + f(\underline{z}_b, b) \left[\frac{1}{2} \sigma_b^2 (\underline{z}_b^2 \phi_b'(\underline{z}_b) + 2 \underline{z}_b \phi_b(\underline{z}_b)) - \mu_{zb}(\underline{z}_b) \phi_b(\underline{z}_b) + \frac{\lambda_b \lambda_B}{\lambda_b + \lambda_B} \right] \\ & - f(\bar{z}_b, b) \left[\frac{1}{2} \sigma_b^2 (\bar{z}_b^2 \phi_b'(\bar{z}_b) + 2 \bar{z}_b \phi_b(\bar{z}_b)) - \mu_{zb}(\bar{z}_b) \phi_b(\bar{z}_b) \right] \\ & = 0. \end{aligned}$$

Since $f(z, B)$ and $f(z, b)$ are arbitrary C^2 functions (only need to satisfy $f'(\underline{z}_B, B) = f'(\bar{z}_B, B) = f'(\underline{z}_b, b) = f'(\bar{z}_b, b) = 0$), we must have that each bracketed term is equal to zero, which implies that Proposition 4. holds after some simplification. $\square$

PROOF OF PROPOSITION 5.. Let τ_i ($i \in \{B, b\}$) be the first regime-switching time and τ_{is} be the first time reaching $\underline{z}_i$. Since the no-transaction region is separated, and upon regime-switching from Regime B to Regime b the investor immediately sells, and thus in Regime B, we have

$$E[\tau_s | z_0 = z] = E[\tau_{Bs} \wedge \tau_B | z_0 = z] = E\left[\int_0^{\tau_{Bs}} e^{-\lambda_B t} dt | z_0 = z\right],$$

and thus Part (i) follows from Proposition 6 in Liu and Loewenstein (2002). In Regime b, upon regime switching from Regime b to Regime B, the investor always buys enough to reach $\bar{z}_B$, and the expected time to the next sale starting from $\bar{z}_B$ is $T_B(\bar{z}_B)$. Thus we have

$$\begin{aligned} E[\tau_s | z_0 = z] &= E[\tau_{bs} \wedge \tau_b | z_0 = z] + T_B(\bar{z}_B) E[1_{\{\tau_b \leq \tau_{bs}\}} | z_0 = z] \\ &= E\left[\int_0^{\tau_{bs}} (1 + \lambda_b T_B(\bar{z}_B)) e^{-\lambda_b t} dt | z_0 = z\right], \end{aligned} \quad (\text{A-71})$$

and thus Part (ii). □

PROOF OF PROPOSITION 6.. Define $W_t \equiv x_t + y_t$ and $\pi_t \equiv \frac{y_t}{x_t + y_t}$. Direct application of Itô's lemma to $e^{-(\nu + \lambda_i)t} C_i(x_t, y_t)$ ($i \in \{B, b\}$) yields that

$$\begin{aligned} & e^{-(\nu + \lambda_i)s} C_i(x_s, y_s) - C_i(x, y) \\ &= \int_0^s e^{-(\nu + \lambda_i)t} \left(\frac{1}{2} \sigma_i^2 z^2 g_i''(z) - \left((\mu_i - r_i)z + \left(\frac{\psi'(z)}{1 - k} \right)^{-1/\gamma} \right) g_i'(z) - (\nu + \lambda_i - \mu_i) g_i(z) \right) y_t dt \\ &+ \int_0^s e^{-(\nu + \lambda_i)t} (g_i(z_t) - (z_t + 1 + \theta_i) g_i'(z_t)) dI_t^* - \int_0^s e^{-(\nu + \lambda_i)t} (g_i(z_t) - (z_t + 1 - \alpha_i) g_i'(z_t)) dD_t^* \end{aligned}$$

$$\begin{aligned}
& + \int_0^s e^{-(\nu+\lambda_i)t} (g_i(z_t) - z_t g_i'(z_t)) \sigma y_t dw_t \\
& = - \int_0^s e^{-(\nu+\lambda_i)t} [\theta_i dI_t^* + \alpha_i dD_t^* + \lambda_i C_j(x_t, y_t) dt] + \int_0^s e^{-(\nu+\lambda_i)t} (g_i(z_t) - z_t g_i'(z_t)) \sigma \pi_t W_t dw_t,
\end{aligned} \tag{A-72}$$

where the second equality follows from (A-4)-(A-6). Then we have

$$dW_t = r_i W_t dt + (\mu_i - r_i) \pi_t W_t dt + \sigma \pi_t W_t dw_t - \alpha_i dD_t^* - \theta_i dI_t^*. \tag{A-73}$$

We then have

$$W_t \leq W_0 e^{\int_0^t (r_i + (\mu_i - r_i) \pi_s) ds} N_t, \tag{A-74}$$

where

$$N_t \equiv e^{-\frac{1}{2} \int_0^t (\pi_s \sigma)^2 ds + \int_0^t \pi_s \sigma dw_s}$$

is a martingale. Thus

$$\begin{aligned}
0 \leq E[e^{-(\nu+\lambda_i)t} C_i(x_t, y_t)] &= E[e^{-(\nu+\lambda_i)t} \pi_t W_t g_i(x_t/y_t)] \\
&\leq E[e^{-(\nu+\lambda_i)t} M W_t] \\
&\leq M E[e^{\int_0^t (r_i + (\mu_i - r_i) \pi_s - \nu - \lambda_i) ds} N_t] \\
&\leq M e^{-(\nu+\lambda_i - r_i - (\mu_i - r_i)/(\underline{z}_B + 1))t} \rightarrow 0, \quad \text{as } t \rightarrow \infty, \tag{A-75}
\end{aligned}$$

where M is a constant, the first inequality follows from the boundedness of the optimal π_t and $g(\cdot)$, the second inequality follows from (A-74), the third inequality holds because the optimal π_t is bounded below by $1/(\underline{z}_B + 1)$ and $E[N_t] = 1$, and the convergence follows from the first assumption in the proposition.

In addition, (A-74) also implies that the last term in (A-72) is a martingale. Therefore,

taking expectation and limit as $s \rightarrow \infty$, and using (A-72) and (A-75), we have that

$$yg_i(x/y) = E \left[\int_0^{\tau_i} e^{-\nu t} (\theta_i dI_t^* + \alpha_i dD_t^*) + e^{-\nu \tau_i} C_j(x_{\tau_i}, y_{\tau_i}) \right].$$

The expressions in (A-7) and (A-8) follow from the fact that in these transaction regions, the investor immediately transacts to the corresponding boundary, incurring the costs represented by the first terms. This completes the proof. $\square$

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